

TIME SERIES ANALYSIS

Complete Study Guide for MSc Statistics Interviews

24 Comprehensive Chapters

150+ Interview Questions | 100+ Worked Examples

50+ Diagrams & Visualizations | Complete Formula Sheets

From Zero to Interview-Ready

ARIMA

ACF/PACF

Stationarity

Forecasting

White Noise

SARIMA

Designed for: Non-Statistics Undergraduates → MSc Statistics Entrance

Covers all concepts from scratch | Simple Language | Visual First Approach

TABLE OF CONTENTS

1	Introduction to Time Series	What, Why, and Where
2	Components of Time Series	Trend, Seasonality, Cyclical, Irregular
3	Time Series Models	Additive vs Multiplicative
4	Time Series Visualization	Plots, Moving Averages, Smoothing
5	Moving Averages	SMA, WMA, CMA — Solved Examples
6	Exponential Smoothing	SES, Holt's, Holt-Winters
7	Stationarity ■	Most Important — Weak/Strong/Strict
8	Differencing	First, Second, Seasonal Differencing
9	Autocorrelation (ACF) ■	Most Important Interview Topic
10	Partial Autocorrelation (PACF)	PACF vs ACF, Graph Reading
11	White Noise	Definition, Properties, Examples
12	Random Walk	Properties, Drift, Stock Prices
13	AR Models	AR(1), AR(2), AR(p) — Full Derivation
14	MA Models	MA(1), MA(2), MA(q)
15	ARMA Models	Combining AR and MA
16	ARIMA ■■■	Most Important — Full p,d,q Guide
17	Seasonal ARIMA (SARIMA)	P,D,Q,s — Seasonal Parameters
18	Model Identification	ACF/PACF Decision Framework
19	Forecasting	One-step, Multi-step, Prediction Intervals
20	Forecast Accuracy	MAE, MSE, RMSE, MAPE
21	Time Series in Statistics	Connections with Regression, Probability
22	Advanced MSc Topics	ADF, KPSS, Cointegration, Unit Root
23	150+ Interview Questions	Conceptual, Numerical, Tricky
24	Complete Cheat Sheet	All Formulas, Definitions, Quick Revision

CHAPTER 1

Introduction to Time Series

Understanding the foundation of temporal data

1.1 What is Time Series?

A **time series** is a sequence of data points collected or recorded at **successive points in time**, usually at **equally spaced intervals**. Think of it like a diary of numbers — every day you write down a number, and together these numbers tell a story that changes over time.

■ Pro Tip

- Simple Definition: Time Series = Data + Time Order
- Key idea: The ORDER of observations MATTERS. Shuffling destroys meaning.
- Each observation depends on what happened BEFORE it — this is what makes it special.

Formal Definition: Let $\{Y_t\} = \{Y_1, Y_2, \dots, Y_T\}$ be a sequence of random variables indexed by time $t = 1, 2, \dots, T$. This sequence is called a **time series**. The subscript t denotes the time period.



1.2 Cross-Sectional vs Time Series Data

To understand time series, let's compare it with the most common type of data you already know — **cross-sectional data**.

Feature	Cross-Sectional Data	Time Series Data
What it is	Data on multiple subjects at ONE point in time	Data on ONE subject at MULTIPLE time points
Order matters?	NO — rows can be shuffled freely	YES — time order is critical
Example	Heights of 100 students in 2024	Daily temperature in Pune, Jan–Dec 2024
Analysis goal	Compare across subjects	Analyse patterns over time

Correlation	Between variables	Between time lags (autocorrelation)
Statistical method	Regression, ANOVA, Clustering	ARIMA, Exponential Smoothing, etc.

1.3 Real-World Examples of Time Series

- **Stock Prices:** NSE NIFTY 50 closing price — recorded every trading day. Pattern: random walk with upward drift over long term.
- ■ **Temperature:** Daily maximum temperature in Pune — shows seasonal pattern (hot summers, cooler winters).
- ■ **Rainfall:** Monthly rainfall in Maharashtra — strong seasonal pattern, low in winter, high in June-September.
- **Population:** India's annual population — shows steady upward trend with no seasonality.
- **GDP:** Quarterly GDP growth — shows trend, business cycles, and seasonal effects.
- **Sales Data:** Monthly sales of umbrellas — obvious seasonal spike every June-August.
- **Heart Rate:** Patient's heart rate every minute — used in healthcare monitoring.
- **Electricity Demand:** Daily power consumption — peaks in mornings and evenings, seasonal patterns.

1.4 Why is Time Series Important?

Time series analysis is one of the most practically important branches of statistics. Here is **why every statistician and data scientist must know it**:

Field	Application	What Time Series Enables
Finance	Stock market prediction, Risk management	Forecast future prices, detect volatility patterns
Economics	GDP forecasting, Inflation analysis	Policy decisions based on economic trends
Healthcare	Epidemic modelling, Patient monitoring	Detect anomalies, predict disease spread
Meteorology	Weather forecasting	Predict rain, temperature, disasters
Supply Chain	Demand forecasting, Inventory management	Reduce overstock and stockouts
Data Science	ML time series models, Feature engineering	Build predictive systems with temporal data
Engineering	Signal processing, Quality control	Detect equipment failures in advance

1.5 Key Vocabulary You Must Know

Term	Meaning	Simple Example
Time Series	Data ordered by time	Daily temperatures
Observation (Y_t)	Data value at time t	Temperature on Day 5 = 32°C
Lag	A previous time point	Yesterday's value = Lag 1

Forecast	Predicted future value	Tomorrow's temperature = 33°C
Residual	Actual minus predicted	32 - 33 = -1 (forecast error)
Stationarity	Statistical properties don't change over time	Constant mean, variance
Trend	Long-term direction (up or down)	Rising GDP over 10 years
Seasonality	Regular pattern that repeats at fixed intervals	Ice cream sales peak every summer

■ Interview Questions

- Q: What is a time series? Give an example. [30-second answer: A sequence of data points collected over time at regular intervals.]
- Q: How is time series data different from cross-sectional data? [Key answer: In time series, ORDER matters. You cannot swap observations.]
- Q: Why can't we use ordinary regression for time series data? [Because observations are NOT independent — each value is influenced by the previous one.]
- Q: Give 3 real-world applications of time series analysis. [Finance: stock prediction; Weather: temperature forecasting; Retail: demand forecasting.]
- Q: What is the difference between a time series and a random variable? [A random variable is a single measurement. A time series is a sequence of measurements over time.]

■ Memory Tricks

- Think of time series as a STORY — the plot matters, and you can't read the chapters out of order!
- T.S. = Time + Sequence. Remember: TIME SEQUENCE.
- Cross-sectional = snapshot (one moment, many people). Time series = movie (one thing, many moments).

CHAPTER 2

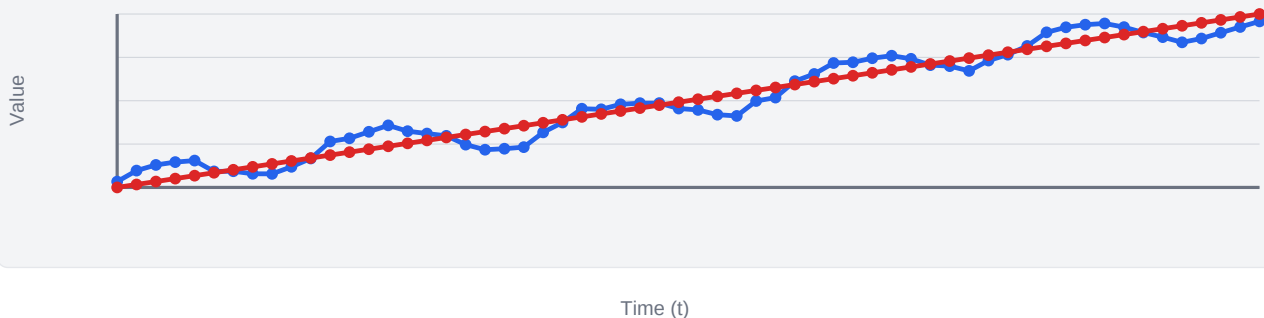
Components of Time Series

Decomposing a time series into its building blocks

2.1 The Four Components — Overview

Every time series can be broken down into **four underlying components**. Understanding these is the **FIRST** step in any time series analysis. Think of it like taking apart a cake to understand its ingredients.

Figure 2.1 — Observed Series with Underlying Trend



The observed series (blue) bounces around the long-term trend (red).

2.2 Trend (T)

Trend is the **long-term movement** or direction in the data. It shows whether the data is generally going **up** (upward trend), **down** (downward trend), or **staying flat** (no trend) over a long period.

Key characteristics of Trend:

- It is **SMOOTH** — short-term fluctuations are ignored.
- It operates over a **LONG** time horizon (years or decades).
- It can be linear (straight line) or non-linear (curved).
- Caused by factors like population growth, economic development, technological change.

Examples of Trends:

Type	Example	Cause
Upward Linear Trend	India's GDP growing steadily	Economic development
Downward Trend	Landline phone usage declining	Replaced by mobile phones
Non-linear Trend	Internet adoption (S-curve growth)	Technology diffusion
No Trend (flat)	Temperature around a constant mean	Stationary data

2.3 Seasonality (S)

Seasonality refers to **regular, predictable fluctuations** that repeat at **fixed intervals** — usually within one calendar year (monthly, quarterly). It is caused by external factors like **weather, holidays, or customs**.



Real-world seasonal examples:

- Umbrella sales: spike every June–August (monsoon season in India)
- Retail sales: spike every October–December (festive season, Diwali, Christmas)
- Ice cream sales: peak in summer months (April–June)
- Agricultural output: depends on sowing and harvest seasons
- Tourism: peak in December (winter holidays in Europe)

2.4 Cyclical Variation (C)

Cyclical variation refers to **fluctuations that occur over longer periods** (usually 2–10 years) and do **NOT** have a **fixed, regular period**. These are caused by **business cycles** — boom, recession, recovery.

Feature	Seasonality	Cyclical Variation
Period	Fixed (e.g., every 12 months)	Variable (2–10+ years)
Regularity	Very regular and predictable	Irregular — period changes each time
Cause	Weather, festivals, customs	Business cycles, economic conditions
Duration	Short (within a year)	Long (multiple years)
Predictability	Highly predictable	Harder to predict exactly
Example	Christmas sales spike every December	Stock market boom and bust cycles

2.5 Irregular/Random Variation (I)

Irregular variation (also called **random** or **residual** variation) is the **unpredictable, erratic fluctuation** remaining after removing trend, seasonality, and cyclical components. It represents **pure randomness** and cannot be explained by any systematic pattern.

- Caused by unexpected events: natural disasters, strikes, pandemics, accidents
- Cannot be predicted or forecasted

- This is what remains after all other components are accounted for
- In a good model, the residuals should look like 'white noise' (pure random)

Figure 2.3 — Pure Irregular/Random Component



No pattern here — this is pure randomness (white noise).

2.6 Summary of All Four Components

Component	Symbol	Nature	Duration	Predictable?	Example
Trend	T	Long-term direction	Years/Decades	Yes	Rising population
Seasonal	S	Regular repeating pattern	< 1 year	Yes	Monsoon sales
Cyclical	C	Wave-like oscillation	2–10 years	Partially	Recession cycles
Irregular	I	Random unpredictable noise	Very short	No	Earthquake impact

■ Interview Questions

- Q: What are the four components of a time series? [Trend (T), Seasonality (S), Cyclical (C), Irregular (I). Always remember all four]
- Q: What is the difference between seasonal and cyclical variation? [Seasonal has FIXED period (within a year, predictable), cyclical has variable period]
- Q: Can a time series have both trend and seasonality? [YES! Most real series have multiple components. E.g., airline passengers over time]
- Q: What does 'decomposition' mean? [It means separating a time series into its T, S, C, I components to study each separately]
- Q: What is irregular variation? Can it be forecasted? [It's the random, unpredictable residual after removing T, S, C. I]

■ Tricky / Trap Questions

- TRAP: 'Seasonal variation always means summer/winter patterns.' WRONG — seasonality just means any REGULAR pattern
- TRAP: 'Cyclical = Seasonal.' WRONG — cyclical has NO fixed period. Seasonal period is fixed. This distinction is critical
- TRAP: 'Irregular variation is just error.' Partially right — it includes measurement error BUT also genuine random shock

■ Memory Tricks

- TSCI = 'The Story Continues Infinitely' — T(rend) S(easonal) C(yclical) I(rregular)
- Seasonal = CALENDAR (fixed timing each year). Cyclical = ECONOMIC (tied to business cycles, variable length).
- Irregular = 'What's left after explaining everything else.' It's the mystery component.

CHAPTER 3

Time Series Models

Additive vs Multiplicative decomposition models

3.1 Why Do We Need Models?

Once we know that a time series has components (T, S, C, I), we need a mathematical way to **combine** these components to represent the observed data. There are **two main ways** to do this:

$$\text{Additive: } Y_t = T_t + S_t + C_t + I_t$$

$$\text{Multiplicative: } Y_t = T_t \times S_t \times C_t \times I_t$$

3.2 Additive Model

In the **additive model**, the four components are simply **added together**. This means each component acts **independently** of the others. The seasonal effect is **constant in absolute terms** — it adds the same amount regardless of the level of the series.

When to use Additive Model:

- The seasonal fluctuations are **ROUGHLY CONSTANT** over time (e.g., sales always increase by exactly 500 units in December).
- The series does **NOT** grow exponentially — changes are **ABSOLUTE** (not percentage-based).
- Seasonal variation is **INDEPENDENT** of the trend level.

Worked Example — Additive Model:

Suppose quarterly sales (units) for Year 1 with:

Trend $T = 200$ (constant), Seasonal effects $S = [+30, -10, +20, -40]$ for Q1–Q4, No cyclical ($C=0$), Irregular $I = [5, -3, 2, -4]$

Quarter	Trend (T)	Seasonal (S)	Irregular (I)	Observed $Y = T+S+I$
Q1	200	+30	+5	$200+30+5 = 235$
Q2	200	-10	-3	$200-10-3 = 187$
Q3	200	+20	+2	$200+20+2 = 222$
Q4	200	-40	-4	$200-40-4 = 156$

3.3 Multiplicative Model

In the **multiplicative model**, the components are **multiplied together**. The seasonal effect is **proportional to the trend level** — so as the series grows, the seasonal swings also grow LARGER. This is more realistic for economic and business data.

When to use Multiplicative Model:

- Seasonal fluctuations INCREASE as the trend increases (e.g., sales increase by 20% every December, not a fixed amount).
- The series grows exponentially or proportionally over time.
- The amplitude of seasonal variation is NOT constant — it grows or shrinks with the series level.

Worked Example — Multiplicative Model:

Quarterly sales with Trend $T = [200, 210, 220, 230]$ and Seasonal indices $S = [1.15, 0.95, 1.10, 0.80]$:

Quarter	Trend (T)	Seasonal Index (S)	Calculated $Y = T \times S$
Q1 (Yr1)	200	1.15	$200 \times 1.15 = 230$
Q2 (Yr1)	210	0.95	$210 \times 0.95 = 199.5$
Q3 (Yr1)	220	1.10	$220 \times 1.10 = 242$
Q4 (Yr1)	230	0.80	$230 \times 0.80 = 184$

Notice: In Q4, the seasonal index is 0.80, meaning sales are 20% BELOW trend. As the trend grows, the absolute drop in Q4 also grows — this is the multiplicative effect.

3.4 Additive vs Multiplicative — Decision Guide

Aspect	Additive Model	Multiplicative Model
Formula	$Y = T + S + C + I$	$Y = T \times S \times C \times I$
Seasonal effect	Constant (fixed units)	Proportional (% of trend)
Look at the graph	Seasonal bands are EQUAL in width	Seasonal bands WIDEN as trend grows
When trend rises...	Seasonal variation stays the same	Seasonal variation grows proportionally
Series type	Linear growth or no growth	Exponential or rapid growth
Real example	Temperature data	Company revenue data
Transformation	No transformation needed	Take log → becomes additive
Seasonal indices	Sum to zero (or near zero)	Multiply to 1 (e.g., average = 1.0)

■ Pro Tip

- KEY TRICK: Plot the data first! If the seasonal 'bumps' get BIGGER as the series grows, use Multiplicative. If bumps stay the same, use Additive.
- Mathematical link: Taking LOG of a Multiplicative model converts it to Additive: $\log(Y) = \log(T) + \log(S) + \log(C) + \log(I)$

■ Interview Questions

- Q: What is the difference between additive and multiplicative time series models? [In additive, $Y=T+S+C+I$ — components
- Q: How do you decide which model to use? [Look at the graph: if seasonal variation is constant → additive; if it grows w
- Q: How can you convert a multiplicative model to an additive model? [Take the NATURAL LOG: $\log(Y) = \log(T) + \log(S) +$
- Q: In the multiplicative model, what does a seasonal index of 1.20 mean? [It means the value for that period is 20% ABOVE
- Q: What is the sum of seasonal indices in an additive model? [They sum to ZERO (over one full seasonal cycle). In multip

■ Common Mistakes

- Saying 'Additive is always better' — WRONG. Choice depends on the data pattern.
- Confusing seasonal INDEX (multiplicative) with seasonal EFFECT (additive). Index is a ratio (e.g., 1.2), effect is an ab
- Forgetting that log transformation converts multiplicative to additive — very common exam question!

CHAPTER 4

Time Series Visualization

Seeing the patterns before calculating anything

4.1 Why Visualize First?

In time series analysis, **plotting your data is the FIRST step** — always, without exception. A good plot tells you: Is there a trend? Is there seasonality? Are there outliers? Is the variance constant? This visual information guides EVERY decision you make later.

4.2 Line Plot (Time Plot)

The **line plot** is the most fundamental time series plot. It plots Y_t (value) on the y-axis against t (time) on the x-axis and connects points with a line.



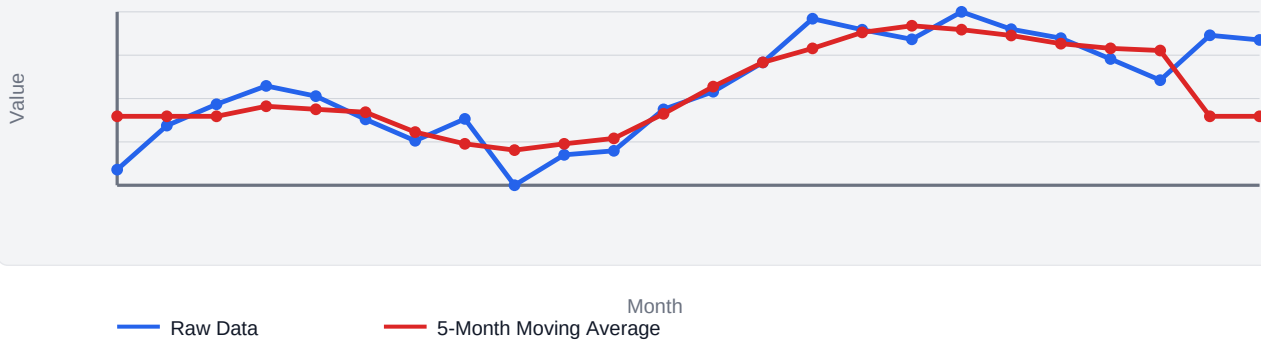
What to look for in a time plot:

What you see	What it means	What to do
General upward/downward direction	TREND present	Use differencing or detrending
Regular repeating peaks/troughs	SEASONALITY present	Apply seasonal adjustment
Spread (variance) increasing	Heteroscedasticity	Log transformation
Sudden jumps or drops	OUTLIERS / structural breaks	Investigate separately
No pattern — random scatter	Possibly stationary / white noise	Check ACF

4.3 Moving Average Smoothing

A **moving average** is a technique to **smooth out short-term fluctuations** and highlight the underlying **trend**. We calculate an average over a 'window' of consecutive observations, then slide that window forward one step at a time.

Figure 4.2 — Raw Data vs 5-Month Moving Average



The red MA line is MUCH smoother — it reveals the trend by filtering out noise.

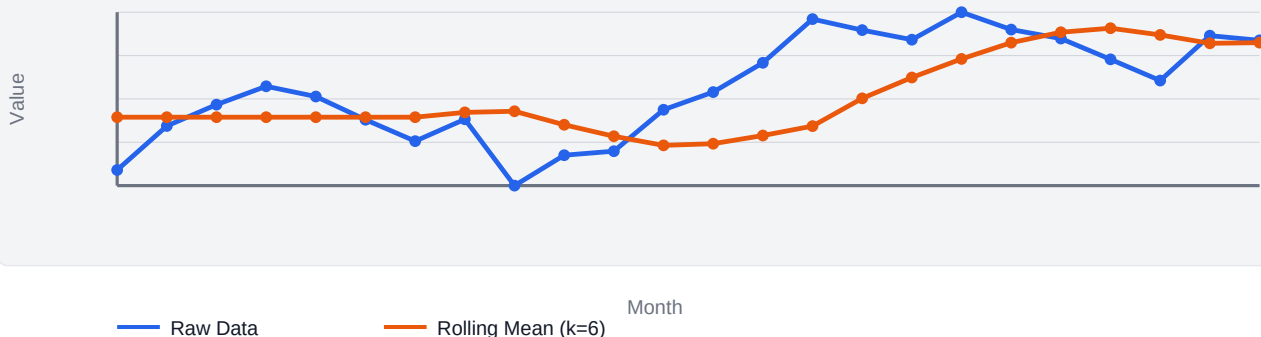
4.4 Rolling Mean and Rolling Variance

Rolling statistics are computed in a moving window and are crucial for checking stationarity:

Rolling Mean: The average of the last k observations. If rolling mean is **approximately constant**, it suggests NO trend (good for stationarity). If it changes over time, trend exists.

Rolling Variance (or Std Dev): The variance of the last k observations. If rolling variance is **approximately constant**, it suggests homoscedasticity (constant spread). If it changes, the series may need a transformation.

Figure 4.3 — Rolling Mean (constant = stationary signal)



If the rolling mean is roughly horizontal, the series is stationary in mean.

4.5 Graph Interpretation for Interviews

Interviewers LOVE to show you a graph and ask 'What do you see?' Here is a systematic framework to answer any graph question:

Step	What to check	Comment out loud
1	Overall direction (trend)	'I can see an upward/downward/no trend...'
2	Regular patterns (seasonality)	'There appears to be a seasonal pattern repeating every...'
3	Spread over time (variance)	'The variance seems constant/increasing/decreasing...'
4	Sudden changes (structural breaks)	'There is a sudden jump around time $t=...$ '

5	Whether it looks stationary	'The series does/does not appear stationary because...'
---	-----------------------------	---

■ Interview Questions

- Q: What is a time plot and what do you look for? [It's a line graph of Y_t vs t . Look for: trend direction, seasonality,
- Q: What does a moving average do to a time series? [It SMOOTHS it — removes short-term noise and highlights the underlying trend]
- Q: If the rolling mean is increasing in a plot, what does it tell you? [The series has a TREND and is likely NON-STATIONARY]
- Q: What is the disadvantage of a moving average? [It loses data points at the ends (no MA for first and last window/2 points)]

CHAPTER 5

Moving Averages

SMA, WMA, CMA — Formulas, Solved Problems, Forecasting

5.1 Simple Moving Average (SMA)

The **Simple Moving Average** of order k at time t is the **arithmetic mean** of the k most recent observations. It gives **equal weight** to all k observations.

$$\text{SMA}(k)_t = (Y_t + Y_{t-1} + \dots + Y_{t-k+1}) / k$$

■ Worked Example — SMA:

Monthly sales (units): 100, 120, 110, 130, 140, 125, 135. Calculate 3-month SMA.

Month (t)	Sales (Y _t)	3-Month SMA = (Y _t + Y _{t-1} + Y _{t-2})/3	Calculation
1	100	—	Not enough data
2	120	—	Not enough data
3	110	110.0	(100+120+110)/3 = 110.0
4	130	120.0	(120+110+130)/3 = 120.0
5	140	126.7	(110+130+140)/3 = 126.7
6	125	131.7	(130+140+125)/3 = 131.7
7	135	133.3	(140+125+135)/3 = 133.3

Interpretation: The SMA at month 3 is 110.0 — this means the average sales over the last 3 months is 110 units. We FORECAST month 8 as the last SMA = 133.3 units.

5.2 Weighted Moving Average (WMA)

The **Weighted Moving Average** assigns **different weights** to observations, with more recent observations typically getting **higher weights**. This is more flexible than SMA because recent data can be made more influential.

$$\text{WMA}_t = (w_1 \cdot Y_t + w_2 \cdot Y_{t-1} + \dots + w_k \cdot Y_{t-k+1}) / (w_1 + w_2 + \dots + w_k)$$

where $w_1 \geq w_2 \geq \dots \geq w_k$ (recent observations get higher weights)

■ Worked Example — WMA:

Monthly sales: 100, 120, 110, 130. Weights: 3 (most recent), 2, 1 (oldest).

Calculate WMA for month 4: $WMA = (3 \times 130 + 2 \times 110 + 1 \times 120) / (3+2+1) = (390+220+120)/6 = 730/6 = \mathbf{121.7}$

Compare: $SMA(3)$ for month 4 = $(120+110+130)/3 = 120.0$

The WMA (121.7) gives more importance to the most recent value of 130, pulling the estimate higher.

5.3 Centered Moving Average (CMA)

The **Centered Moving Average** is used when the order k is **even** (e.g., $k=4$ or $k=12$). For odd k , the MA naturally centers at the middle observation. For even k , we take a second moving average of the moving averages to center it.

CMA: Average of two consecutive 4-point MAs → centered at integer time point

■ Worked Example — CMA (order 4):

Quarterly data: $Q1=40, Q2=55, Q3=65, Q4=48, Q1=42, Q2=58, Q3=70, Q4=50$

Quarter (t)	Value	4-pt MA	Centered MA (CMA)
Q1 (t=1)	40	—	—
Q2 (t=2)	55	—	—
Q3 (t=3)	65	$(40+55+65+48)/4 = 52.0$	$(52.0+52.5)/2 = 52.25$
Q4 (t=4)	48	$(55+65+48+42)/4 = 52.5$	$(52.5+53.75)/2 = 53.125$
Q1 (t=5)	42	$(65+48+42+58)/4 = 53.25$	$(53.25+55.5)/2 = 54.375$
Q2 (t=6)	58	$(48+42+58+70)/4 = 54.5$	$(54.5+55.0)/2 = 54.75$
Q3 (t=7)	70	$(42+58+70+50)/4 = 55.0$	—
Q4 (t=8)	50	—	—

■ Pro Tip

- CMA is used in the DECOMPOSITION method to extract the trend-cycle component from quarterly or monthly data.
- Why center? Because a 4-point MA would fall BETWEEN time points (e.g., between Q2 and Q3). Centering aligns it with a time point.

5.4 Advantages and Disadvantages

Type	Advantages	Disadvantages
SMA	Simple to calculate; good for stable series	Lags behind actual data; equal weight is unrealistic; loses end data
WMA	More flexible; recent data gets more weight	Choosing weights is subjective; more complex
CMA	Best for seasonal decomposition; properly centered	Cannot compute at the extremes; requires 2 passes

5.5 Using MA for Forecasting

For forecasting, the **SMA is a naive but useful method**: The forecast for the next period is simply the current SMA.

$$\text{Forecast: } \hat{Y}_{t+1} = \text{SMA}(k)_t = (Y_t + Y_{t-1} + \dots + Y_{t-k+1}) / k$$

■ Interview Questions

- Q: What is a moving average? [An average calculated over a sliding window of k observations, used to smooth data and remove noise.]
- Q: What is the difference between SMA and WMA? [SMA gives equal weights to all k observations. WMA assigns higher weights to more recent observations.]
- Q: Why do we use a Centered Moving Average? [When order k is even, the simple MA falls between time points. Centering the MA aligns it with the time points.]
- Q: What is the main weakness of moving averages for forecasting? [They LAG behind the actual data. In a trending series, the forecast will be lower than the actual data.]
- Q: How many data points are 'lost' in a 5-period moving average? [2 at the beginning and 2 at the end = 4 total lost. Generally, k/2 points are lost at each end.]

■ Common Mistakes

- Forgetting to center the MA when order is EVEN. Always use CMA for even-order MAs in decomposition.
- Using MA to forecast a seasonal series without adjusting for seasonality — MA smooths out seasonality, so the forecast will be lower than the actual data.
- Choosing too large a window: over-smoothing hides useful patterns. Too small: doesn't remove enough noise.

CHAPTER 6

Exponential Smoothing

SES, Double ES, Holt's, and Holt-Winters Methods

6.1 Why Exponential Smoothing?

Moving averages have a major flaw: they give **equal weight** to old and recent data. In reality, **recent observations are more relevant**. Exponential Smoothing fixes this by giving **exponentially decreasing weights** to past observations — recent data gets the most weight, older data gets less and less.

6.2 Simple Exponential Smoothing (SES)

SES is used for series with **no trend** and **no seasonality**. It produces a smoothed value S_t which is also the one-step-ahead forecast.

$$S_t = \alpha \cdot Y_t + (1-\alpha) \cdot S_{t-1}$$

where α (**alpha**) is the **smoothing parameter**, $0 < \alpha < 1$.

Understanding α (Alpha):

Value of α	Effect	When to use
α close to 1 (e.g., 0.9)	High weight on recent data, low memory	Rapidly changing series, recent data is most informative
α close to 0 (e.g., 0.1)	Low weight on recent data, high memory	Stable series, want smooth output, past data still relevant
$\alpha = 0.5$	Equal balance between old and new	Moderately changing series

■ Worked Example — SES with $\alpha = 0.3$:

Sales data: 50, 55, 58, 62, 60. Initial $S_1 = Y_1 = 50$.

t	Y_t	$S_t = 0.3 \cdot Y_t + 0.7 \cdot S_{t-1}$	Forecast $\hat{Y}_{t+1} = S_t$
1	50	50.0 (initialise)	50.0
2	55	$0.3 \times 55 + 0.7 \times 50 = 16.5 + 35 = 51.5$	51.5
3	58	$0.3 \times 58 + 0.7 \times 51.5 = 17.4 + 36.05 = 53.45$	53.45
4	62	$0.3 \times 62 + 0.7 \times 53.45 = 18.6 + 37.42 = 56.02$	56.02
5	60	$0.3 \times 60 + 0.7 \times 56.02 = 18 + 39.21 = 57.21$	57.21
6 (Forecast)	—	—	Forecast = 57.21

6.3 Double Exponential Smoothing / Holt's Method

SES cannot handle a **trend**. **Holt's Method** (Double Exponential Smoothing) adds a second equation to track the trend. It uses **two parameters**: α and β .

$$\text{Level: } L_t = \alpha \cdot Y_t + (1-\alpha) \cdot (L_{t-1} + T_{t-1})$$

$$\text{Trend: } T_t = \beta \cdot (L_t - L_{t-1}) + (1-\beta) \cdot T_{t-1}$$

$$\text{Forecast } h \text{ steps ahead: } \hat{Y}_{t+h} = L_t + h \cdot T_t$$

where β (**beta**) is the trend smoothing parameter ($0 < \beta < 1$). L_t is the estimated level, T_t is the estimated trend.

■ Worked Example — Holt's Method ($\alpha=0.4$, $\beta=0.3$):

Sales: 200, 210, 225, 240. Initialize: $L_1=200$, $T_1=10$.

t	Y_t	L_t	T_t	1-step Forecast
1	200	200.0 (init)	10.0 (init)	—
2	210	$0.4 \times 210 + 0.6 \times (200 + 10) = 84 + 126 = 210.0$	$0.3 \times (210 - 200) + 0.7 \times 10 = 3 + 7 = 10.0$	$200 + 1 \times 10 = 210.0$
3	225	$0.4 \times 225 + 0.6 \times (210 + 10) = 90 + 132 = 222.0$	$0.3 \times (222 - 210) + 0.7 \times 10 = 3.6 + 7 = 10.6$	$210 + 1 \times 10 = 220.0$
4	240	$0.4 \times 240 + 0.6 \times (222 + 10.6) = 96 + 139.6 = 235.6$	$0.3 \times (235.6 - 222) + 0.7 \times 10.6 = 4.08 + 7.42 = 11.5$	$222 + 1 \times 10.6 = 232.6$
5 (Forecast)	—	—	—	$235.6 + 1 \times 11.5 = 247.1$

6.4 Holt-Winters Method (Triple Exponential Smoothing)

For series with **both trend AND seasonality**, we use the **Holt-Winters method**, which adds a third equation for seasonality. Uses **three parameters**: α , β , γ .

Parameter	Symbol	Controls	Effect when close to 1	Effect when close to 0
Alpha	α	Level smoothing	High responsiveness to recent Y_t	More weight on past level
Beta	β	Trend smoothing	Trend adapts quickly to new data	Trend is very stable/slow to change
Gamma	γ	Seasonal smoothing	Seasonal indices update quickly	Seasonal indices are very stable

$$\text{Additive HW: } \hat{Y}_{t+h} = (L_t + h \cdot T_t) + S_{t+h-m}$$

$$\text{Multiplicative HW: } \hat{Y}_{t+h} = (L_t + h \cdot T_t) \times S_{t+h-m}$$

where m = length of seasonal cycle (e.g., $m=12$ for monthly, $m=4$ for quarterly)

6.5 Choosing Between SES, Holt's, and Holt-Winters

Method	Trend?	Seasonality?	Parameters	Use Case
Simple ES (SES)	No	No	α only	Stable series around a constant level
Holt's Method	Yes	No	α, β	Series with trend but no seasonality
Holt-Winters (Additive)	Yes	Yes (constant amplitude)	α, β, γ	Series with both, constant seasonal swings
Holt-Winters (Multiplicative)	Yes	Yes (growing amplitude)	α, β, γ	Series with growing seasonal swings

■ Interview Questions

- Q: What is exponential smoothing? [A forecasting method that gives exponentially decreasing weights to past observations]
- Q: What does the parameter α control in SES? [It controls the balance between the current observation and past smoother]
- Q: What is the difference between SES and Holt's Method? [SES handles only stationary series (no trend). Holt's adds a trend component]
- Q: What does γ (gamma) control in Holt-Winters? [It controls how quickly the seasonal indices are updated. High γ = seasonal changes are updated quickly]
- Q: If $\alpha = 1$ in SES, what happens? [The forecast for the next period is just the LAST observed value (naive forecast). $S_t = y_t$]

■ Memory Tricks

- SES = No trend, no season (ONE component smoothed, ONE parameter α)
- Holt's = Trend only (TWO components smoothed, TWO parameters α, β)
- Holt-Winters = Trend + Season (THREE components, THREE parameters α, β, γ)
- Acronym: 'Some Hot Women' = SES, Holt, Holt-Winters ■

CHAPTER 7

Stationarity ■ MOST IMPORTANT

The foundational concept every interviewer will test you on

■ WHY THIS CHAPTER IS #1

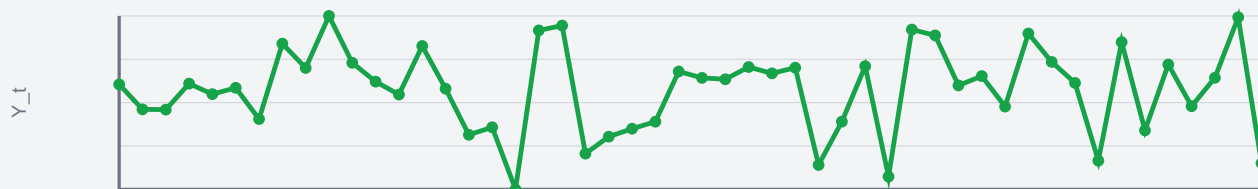
- Stationarity is the **SINGLE** most tested concept in MSc Statistics interviews.
- Almost every model (AR, MA, ARIMA) **REQUIRES** the series to be stationary.
- Understanding stationarity separates a prepared candidate from an unprepared one.
- Every interviewer will ask: 'What is stationarity?' — your answer must be flawless.

7.1 What is Stationarity? — Intuition First

A time series is **stationary** if its **statistical properties do not change over time**. Imagine you're looking at the series through a window. If you slide that window to any position in time, the **mean, variance, and correlation structure** you see always look the same. The series is 'well-behaved' — it fluctuates around a fixed level with consistent spread.

Simple Analogy: A stationary series is like a **calm river** — the water level stays roughly constant over time. A non-stationary series is like a river that gradually floods — the level keeps rising (trend) or the floods get worse and worse (increasing variance).

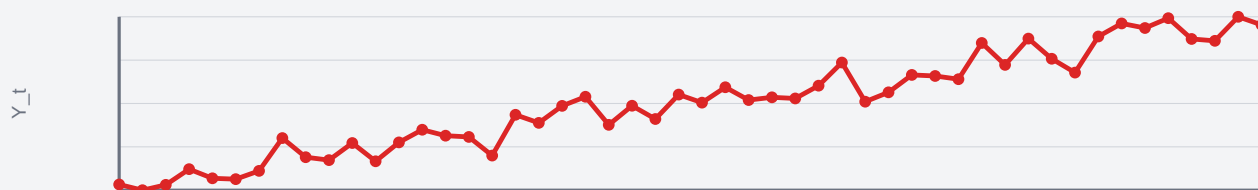
Figure 7.1 — **STATIONARY**: Constant mean (~50), constant variance



— Stationary Series

■ Stationary — flat rolling mean, consistent spread throughout.

Figure 7.2 — **NON-STATIONARY**: Rising mean (trend present)



— Non-stationary (Trend)

■ Not stationary — the mean is increasing over time (trend).

7.2 Formal Definition — Weak (Covariance) Stationarity

In practice, most statistical models require **Weak Stationarity** (also called **Second-Order Stationarity** or **Covariance Stationarity**). This is defined by THREE conditions:

Condition 1: $E[Y_t] = \mu$ (constant mean for all t)

Condition 2: $\text{Var}(Y_t) = \sigma^2$ (constant, finite variance for all t)

Condition 3: $\text{Cov}(Y_t, Y_{t+k}) = \gamma_k$ (depends ONLY on lag k , not on t)

Breaking this down in plain English:

- Condition 1: The average value of the series doesn't change over time. No trend.
- Condition 2: The spread (variance) of the series doesn't change over time. No heteroscedasticity.
- Condition 3: The correlation between Y_t and Y_{t+k} depends only on how far apart they are (lag k), not on when in time you measure it.

7.3 Strict (Strong) Stationarity

Strict Stationarity is a much stronger condition. It requires that the **entire joint probability distribution** of $(Y_{t1}, Y_{t2}, \dots, Y_{tk})$ is the same as $(Y_{t1+h}, Y_{t2+h}, \dots, Y_{tk+h})$ for any shift h . This means ALL statistical properties (not just mean and variance) are time-invariant.

Feature	Weak (Covariance) Stationarity	Strict (Strong) Stationarity
Requirements	Constant mean, variance, and autocovariance	Entire joint distribution is time-invariant
How strong?	Weaker condition	Much stronger condition
Does A imply B?	Weak does NOT imply strict in general	Strict IMPLIES weak (usually)
Exception	—	For Gaussian processes, weak = strict
Practical use	Used in almost all time series models	Rarely required in practice
Easier to test?	YES — test mean, variance, ACF	Very hard to test in practice

■ Pro Tip

- In almost ALL interviews, when they say 'stationarity' they mean WEAK stationarity.
- The key question: Does strict stationarity imply weak stationarity? ANSWER: YES (with finite variance). Does weak imply

7.4 Examples of Stationary and Non-Stationary Series

Series	Stationary?	Reason	Fix
White noise: $\varepsilon_t \sim \text{iid}(0, \sigma^2)$	YES ■	Mean=0, Var= σ^2 both constant, no autocorrelation	Already stationary
Random walk: $Y_t = Y_{t-1} + \varepsilon_t$	NO ■	Variance grows with t: $\text{Var}(Y_t) = t\sigma^2$ (not constant)	First difference
Trending series (e.g., GDP)	NO ■	Mean increases over time	Differencing or detrending
Seasonal series (sales)	NO ■	Mean changes with season (periodic)	Seasonal differencing
AR(1) with $ \phi < 1$	YES ■	Stable: mean=0, variance= $\sigma^2/(1-\phi^2)$	Already stationary
AR(1) with $\phi=1$ (unit root)	NO ■	This IS a random walk — variance is infinite	First difference
Stationary around a trend	NO ■	Mean changes over time (deterministic trend)	Detrend by regression

7.5 How to Make a Series Stationary

Step-by-step approach to achieve stationarity:

Step	Problem detected	Transformation to apply
1	Upward/downward trend in mean	Apply FIRST DIFFERENCING: $\Delta Y_t = Y_t - Y_{t-1}$
2	Trend still present after 1st diff	Apply SECOND DIFFERENCING: $\Delta^2 Y_t = \Delta Y_t - \Delta Y_{t-1}$
3	Seasonal pattern present	Apply SEASONAL DIFFERENCING: $\Delta_s Y_t = Y_t - Y_{t-s}$
4	Variance is not constant (increases)	Apply LOG TRANSFORMATION: $\log(Y_t)$ or $\sqrt{Y_t}$
5	Still non-stationary	Combine: log + differencing

7.6 Testing for Stationarity

Method	Type	How it works	Null Hypothesis
Visual inspection	Informal	Plot series, rolling mean, rolling variance	No formal null — use judgment
ACF plot	Informal	If ACF decays slowly → non-stationary	No formal null
ADF Test	Formal	Augmented Dickey-Fuller unit root test	H0: Series has unit root (NON-stationary)
KPSS Test	Formal	Kwiatkowski-Phillips-Schmidt-Shin test	H0: Series IS stationary
PP Test	Formal	Phillips-Perron test (similar to ADF)	H0: Series has unit root (NON-stationary)

■ Pro Tip

- ADF test: REJECT H0 → series IS stationary ($p < 0.05$ means stationary). Remember: rejecting H0 is GOOD here!
- KPSS test: FAIL TO REJECT H0 → series IS stationary. The null hypotheses are OPPOSITE to ADF.
- Best practice: Use BOTH ADF and KPSS together. If ADF rejects AND KPSS doesn't reject → strong evidence of stationarity.

■ Interview Questions

- Q: What is stationarity? [A time series is weakly stationary if its mean, variance, and autocovariance structure are all
- Q: What are the three conditions for weak stationarity? [1) $E[Y_t] = \mu$ (constant mean), 2) $\text{Var}(Y_t) = \sigma^2$ (constant variance)
- Q: Why does stationarity matter? [Most time series models (AR, MA, ARIMA) assume stationarity. A non-stationary series fl
- Q: What is the difference between weak and strict stationarity? [Weak requires only constant mean, variance, autocovaria
- Q: Is white noise stationary? [YES. White noise has mean=0, constant variance σ^2 , and zero autocorrelation at all lags —
- Q: Is a random walk stationary? [NO. $Y_t = Y_{t-1} + \varepsilon_t$ has $\text{Var}(Y_t) = t\sigma^2$, which grows with time. The variance is NOT

■ Tricky / Trap Questions

- TRAP: 'If a series fluctuates, it's stationary.' WRONG — fluctuation alone doesn't mean stationary. A seasonal series fl
- TRAP: 'Differencing always fixes non-stationarity.' Not always! If non-stationarity is caused by a DETERMINISTIC trend (
- TRAP: 'Strict stationarity is what we use in practice.' WRONG — we almost always mean WEAK (covariance) stationarity.

■ Memory Tricks

- Stationarity = TIME-INVARIANT statistics. Imagine taking a photo of the series at any point — it always looks the same.
- Three conditions: MVC — Mean constant, Variance constant, Covariance depends only on lag.
- Easy test: Plot the rolling mean. If it's flat → stationary in mean. Plot rolling std dev. If flat → stationary in varia

CHAPTER 8

Differencing

The most powerful tool for achieving stationarity

8.1 What is Differencing?

Differencing is the process of **computing the difference between consecutive observations**. It is the most common and effective way to remove trend and achieve stationarity. Think of it as asking: 'By how much did the value CHANGE from one period to the next?'

$$\text{First Difference: } \Delta Y_t = Y_t - Y_{t-1}$$

8.2 First Differencing

Why it works: If $Y_t = T_t + \varepsilon_t$ where T_t is a linear trend, then:

$$\Delta Y_t = Y_t - Y_{t-1} = (T_t + \varepsilon_t) - (T_{t-1} + \varepsilon_{t-1}) = (T_t - T_{t-1}) + (\varepsilon_t - \varepsilon_{t-1})$$

If the trend is LINEAR, $T_t - T_{t-1} = \text{constant}$. So differencing removes the linear trend!

Figure 8.1 — Original Non-Stationary Series (trending upward)

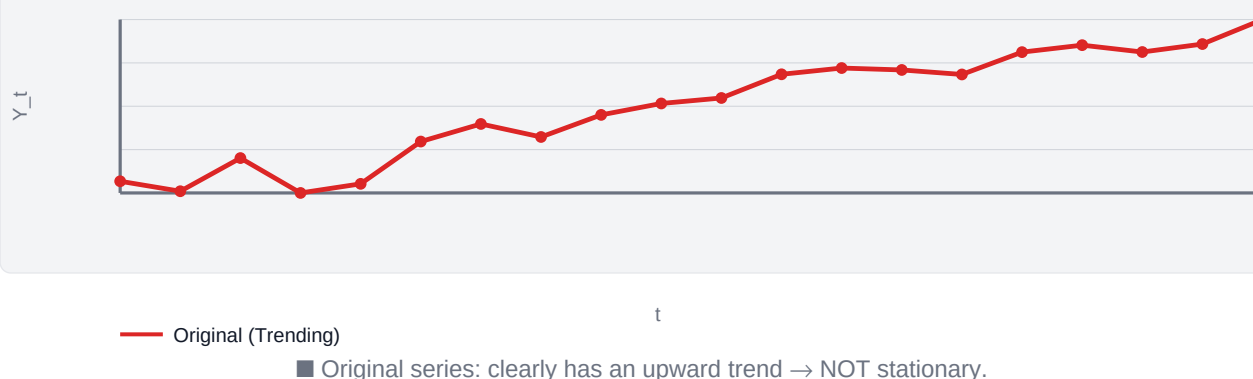
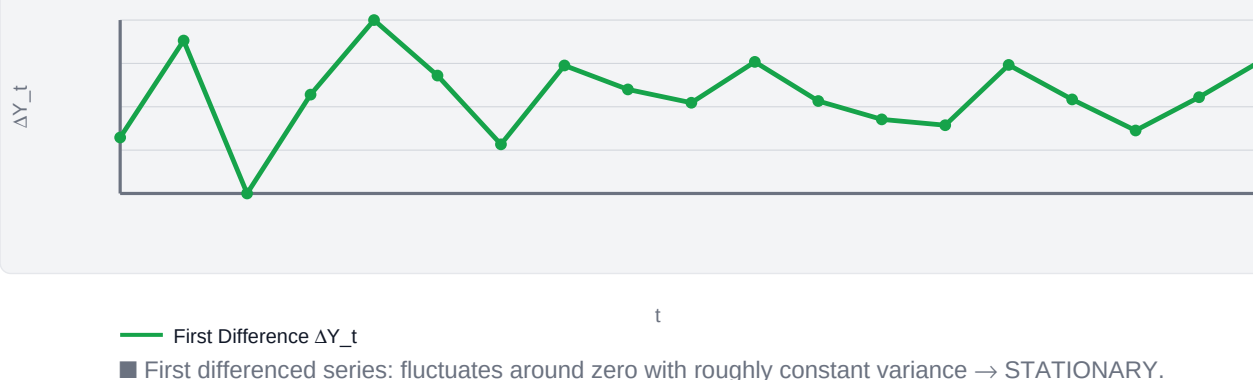


Figure 8.2 — After First Differencing (trend removed)



■ Numerical Example — First Differencing:

GDP (£ trillion): 80, 83, 87, 90, 94, 98

Year (t)	GDP Y_t	First Difference $\Delta Y_t = Y_t - Y_{t-1}$	Interpretation
1	80	—	No previous value
2	83	$83 - 80 = 3$	+3 trillion change
3	87	$87 - 83 = 4$	+4 trillion change
4	90	$90 - 87 = 3$	+3 trillion change
5	94	$94 - 90 = 4$	+4 trillion change
6	98	$98 - 94 = 4$	+4 trillion change

The differenced series {3, 4, 3, 4, 4} is roughly constant (stationary) — it removed the linear trend!

8.3 Second Differencing

If first differencing does not achieve stationarity (e.g., there is a **quadratic trend**), we apply **second differencing**:

Second Difference: $\Delta^2 Y_t = \Delta Y_t - \Delta Y_{t-1} = Y_t - 2Y_{t-1} + Y_{t-2}$

Warning: Rarely go beyond second differencing. Over-differencing introduces artificial patterns and makes things worse. In practice, $d=1$ is most common, $d=2$ is sometimes needed, $d \geq 3$ is almost never used.

8.4 Seasonal Differencing

Seasonal differencing removes seasonality by subtracting the value from the **same season in the previous year**. For monthly data ($s=12$) or quarterly data ($s=4$):

Seasonal Difference (period s): $\Delta_s Y_t = Y_t - Y_{t-s}$

Example: Monthly umbrella sales. June 2024 minus June 2023 = change in June sales year-over-year. This removes the June 'seasonal spike'.

■ **Seasonal Differencing Example ($s=4$, quarterly):**

Quarter	Sales Y_t	Seasonal Diff $\Delta_4 Y_t = Y_t - Y_{t-4}$	Meaning
Q1 2023	100	—	Base year
Q2 2023	130	—	Base year
Q3 2023	150	—	Base year
Q4 2023	90	—	Base year
Q1 2024	108	$108 - 100 = 8$	Q1 grew by 8 YoY

Q2 2024	140	$140 - 130 = 10$	Q2 grew by 10 YoY
Q3 2024	162	$162 - 150 = 12$	Q3 grew by 12 YoY
Q4 2024	95	$95 - 90 = 5$	Q4 grew by 5 YoY

8.5 How Many Differences? — Choosing 'd'

d value	Meaning	When to use
d = 0	No differencing needed	Series is already stationary
d = 1	First differencing once	Series has a linear trend (most common)
d = 2	Second differencing	Series has a quadratic trend (rare)
d = 3+	Higher order differencing	Almost never — if needed, suspect a mistake

■ Interview Questions

- Q: What is differencing and why do we use it? [Differencing computes $Y_t - Y_{t-1}$ to remove trend. It transforms a non-stationary series into a stationary one.]
- Q: What is the formula for second differencing? [$\Delta^2 Y_t = Y_t - 2Y_{t-1} + Y_{t-2}$. It removes a quadratic trend.]
- Q: What is seasonal differencing? [$\Delta_s Y_t = Y_t - Y_{t-s}$. Subtract the value from the same period s steps ago to remove seasonality.]
- Q: Why shouldn't we over-difference? [Over-differencing introduces artificial negative autocorrelation. It makes the ACF plot more complex and harder to interpret.]
- Q: If $d=1$ in $ARIMA(1,1,1)$, what does that mean? [The series was differenced ONCE to achieve stationarity before fitting the AR and MA components.]

■ Memory Tricks

- $d=1$ removes LINEAR trend. $d=2$ removes QUADRATIC trend. Just like differentiation in calculus!
- $s=12$ for monthly data (one year = 12 months). $s=4$ for quarterly. $s=7$ for daily with weekly seasonality.
- Rule of thumb: If rolling mean drifts, use $d=1$. If variance grows, use log first.

CHAPTER 9

Autocorrelation (ACF) ■ Most Important Interview Topic

The heartbeat of time series analysis — master this completely

■ Why ACF is the #1 Interview Topic

- ACF plots are shown in almost EVERY time series interview.
- ACF determines whether to use AR or MA models.
- ACF tells you if a series is stationary.
- Understanding ACF is prerequisite for ARMA and ARIMA.
- Every follow-up question about model selection involves ACF.

9.1 What is Autocorrelation?

Autocorrelation is the **correlation of a time series with its own past values**. Just as regular correlation measures the relationship between TWO different variables, autocorrelation measures the relationship between Y_t and Y_{t-k} (the same variable, but k time steps apart).

Simple intuition: If today's temperature is high, is tomorrow's temperature likely to also be high? If yes, there is **positive autocorrelation at lag 1**. This is exactly what autocorrelation measures.

9.2 Mathematical Definition of Autocorrelation

The **autocovariance** at lag k is defined as:

$$\gamma_k = \text{Cov}(Y_t, Y_{t-k}) = E[(Y_t - \mu)(Y_{t-k} - \mu)]$$

The **autocorrelation coefficient** at lag k is the autocovariance normalized by the variance:

$$\rho_k = \gamma_k / \gamma_0 = \text{Cov}(Y_t, Y_{t-k}) / \text{Var}(Y_t)$$

where $\gamma_0 = \text{Var}(Y_t)$ is the variance (autocovariance at lag 0). Note: $\rho_0 = 1$ always.

Range: $-1 \leq \rho_k \leq 1$. Just like regular correlation!

9.3 Sample Autocorrelation Function (ACF) — Formula

In practice, we estimate the ACF from data:

$$r_k = \sum (Y_t - \bar{Y})(Y_{t-k} - \bar{Y}) / \sum (Y_t - \bar{Y})^2 \text{ for } t = k+1 \text{ to } T$$

where $\bar{Y}$ is the sample mean and the sums run from $t=k+1$ to T for the numerator.

■ Worked Example — Calculating ACF at Lag 1:

Data: $Y = \{4, 6, 5, 8, 7\}$. Mean $\bar{Y} = (4+6+5+8+7)/5 = 6.0$

t	Y_t	$Y_t - \bar{Y}$	Y_{t-1}	$Y_{t-1} - \bar{Y}$	$(Y_t - \bar{Y})(Y_{t-1} - \bar{Y})$	$(Y_t - \bar{Y})^2$
2	6	0	4	-2	$0 \times (-2) = 0$	0
3	5	-1	6	0	$-1 \times 0 = 0$	1
4	8	2	5	-1	$2 \times (-1) = -2$	4
5	7	1	8	2	$1 \times 2 = 2$	1
Sum	—	—	—	—	$0+0-2+2 = 0$	$0+1+4+1=6$

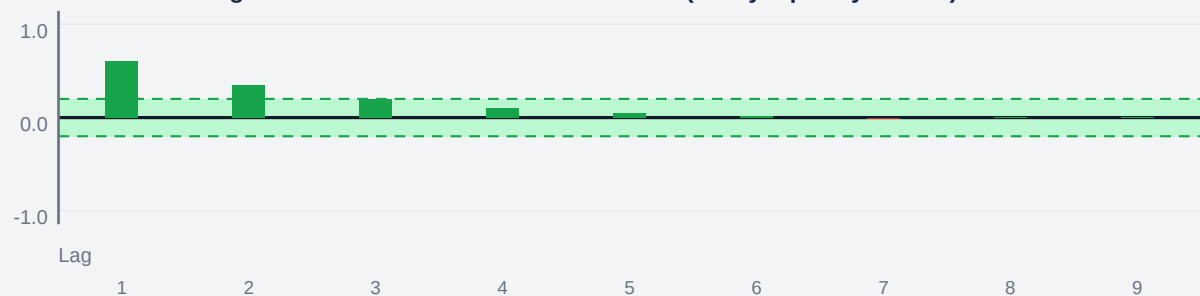
$$r_1 = 0 / 6 = 0.0$$

Interpretation: There is NO autocorrelation at lag 1 for this data.

9.4 The ACF Plot — Correlogram

The **Autocorrelation Function (ACF) plot** (also called a **correlogram**) displays ρ_k (y-axis) against lag k (x-axis). It is one of the most important diagnostic tools in time series analysis.

Figure 9.1 — ACF of STATIONARY Series (decays quickly to zero)



■ STATIONARY: ACF drops off quickly (geometric decay) and stays within confidence bands.

Figure 9.2 — ACF of NON-STATIONARY Series (slow decay — red flag!)



■ NON-STATIONARY: ACF is large and decays VERY slowly — classic sign of non-stationarity.

9.5 Confidence Bands (Significance Lines)

In an ACF plot, the **dashed blue lines** are the **95% confidence bounds**. If an autocorrelation bar **extends beyond** these lines, it is **statistically significant** (i.e., significantly different from zero). The bounds are approximately:

$$\text{95\% Confidence Bounds: } \pm 1.96 / \sqrt{T}$$

For $T=100$ observations: $\pm 1.96 / \sqrt{100} = \pm 1.96 / 10 = \pm 0.196$. So any $|r_k| > 0.196$ is significant.

9.6 ACF Patterns and What They Tell You

ACF Pattern	What it indicates	Action
Decays exponentially (geometric)	AR process (stationary)	AR model, order = where it cuts off
Cuts off sharply after lag q	MA(q) process	Use MA(q) model
Slow, linear decay (all lags large)	NON-STATIONARY — unit root	Apply differencing first!
Significant spikes at lags 1, s , $2s$...	SEASONAL pattern at period s	Apply seasonal differencing
All near zero (within bands)	White noise — no pattern	No model needed
Oscillating (alternating + and -)	Possible negative AR or overdifferencing	Investigate MA or reduce d
Large spike at lag 1 only	MA(1) process	Use MA(1) model

9.7 Positive vs Negative Autocorrelation

Type	ρ_k value	Meaning	Real Example
Positive Autocorrelation	$\rho_k > 0$	High values tend to follow high values (smoothness)	Temperature: hot day followed by hot day
Negative Autocorrelation	$\rho_k < 0$	High values tend to follow low values (oscillation)	Economic series that 'bounce back'
Zero Autocorrelation	$\rho_k = 0$	No relationship between Y_t and Y_{t-k}	White noise: completely random

Interview Questions

- Q: What is autocorrelation? [Autocorrelation is the correlation of a time series with its own lagged values. ρ_k measure
- Q: What is the ACF and how do you interpret it? [The ACF plot shows ρ_k vs lag k . Bars beyond the dashed confidence b
- Q: What is ρ_0 (autocorrelation at lag 0)? [ρ_0 is always 1.0. It's the correlation of Y_t with itself, which is always
- Q: If the ACF decays slowly, what should you do? [The series is likely NON-STATIONARY. Apply first differencing and the
- Q: What are the confidence bands in an ACF plot? [They are $\pm 1.96 / \sqrt{T}$ — the 95% significance bounds. Spikes beyond the
- Q: Describe the ACF pattern of an MA(2) process. [The ACF has significant spikes at lags 1 and 2, then CUTS OFF sharp

■ ■ Tricky / Trap Questions

- TRAP: 'High ACF at lag 1 means the series is AR(1).' NOT NECESSARILY — it could be part of any AR(p) or even a non-
- TRAP: 'If ACF cuts off after lag 2, use AR(2).' WRONG — if ACF cuts off, it's MA(2). AR models show TAILING OFF in ACF
- TRAP: 'Lag 0 is plotted in the ACF.' It is always 1.0 — some plots show it, some don't. Don't be confused if it's missing

■ Memory Tricks

- ACF CUT OFF after lag $q \rightarrow$ MA(q). ACF TAILS OFF $\rightarrow$ AR process.
- Slow ACF decay = non-stationary. Fast decay = stationary. This is THE most important visual rule.
- $\rho_k = \gamma_k / \gamma_0$. Think: normalize autocovariance by variance, just like regular correlation = Cov/StdDev.

CHAPTER 10

Partial Autocorrelation (PACF)

The companion to ACF — completing the AR vs MA identification toolkit

10.1 Why PACF? The Problem with ACF Alone

The ACF at lag k captures the correlation between Y_t and Y_{t-k} , but this includes **indirect effects** through all the intermediate lags (lag 1, 2, ..., $k-1$). The ACF at lag 3, for example, partly reflects the correlations at lags 1 and 2 'passing through'.

PACF solves this: The **Partial Autocorrelation Function (PACF)** at lag k measures the correlation between Y_t and Y_{t-k} **after removing the linear effects of all intermediate lags** ($Y_{t-1}, \dots, Y_{t-k+1}$).

Simple analogy: Imagine you want to know if a student's score in Week 3 is directly influenced by Week 1. But Weeks 1 and 2 are correlated, and Week 2 and 3 are correlated. PACF finds the **DIRECT** connection between Week 1 and Week 3, removing Week 2's influence.

10.2 Mathematical Definition of PACF

The PACF at lag k , denoted ϕ_{kk} , is the last coefficient in an AR(k) regression:

$$Y_t = \phi_{k1}Y_{t-1} + \phi_{k2}Y_{t-2} + \dots + \phi_{kk}Y_{t-k} + \text{error}$$

The **PACF at lag $k = \phi_{kk}$** — the coefficient of Y_{t-k} in the above regression. It represents the **DIRECT** effect of Y_{t-k} on Y_t after controlling for all shorter lags.

PACF at lag 1 = ACF at lag 1 (because there are no intermediate lags to control for)

$$\phi_{11} = \rho_1 \text{ (PACF lag 1 = ACF lag 1)}$$

$$\phi_{22} = (\rho_2 - \rho_1^2) / (1 - \rho_1^2) \text{ (Yule-Walker equation for PACF lag 2)}$$

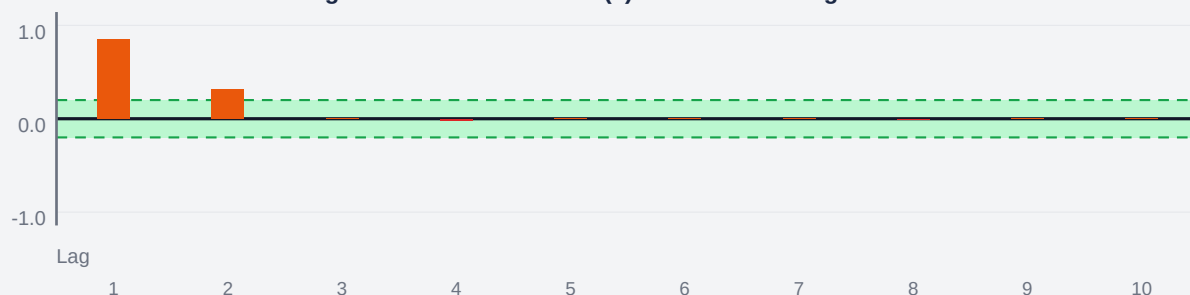
10.3 PACF Patterns and What They Tell You

Figure 10.1 — ACF of AR(2): Tails off gradually



ACF of AR(2): decays gradually (tails off) — does NOT cut off sharply.

Figure 10.2 — PACF of AR(2): Cuts off after lag 2



PACF of AR(2): SIGNIFICANT at lags 1 and 2, then CUTS OFF sharply → order $p=2$ identified!

10.4 The Golden Rule: ACF + PACF Table

Model	ACF Behaviour	PACF Behaviour	How to identify
AR(p)	Tails off (exponential decay or oscillation)	CUTS OFF after lag p	PACF: count significant spikes → order p
MA(q)	CUTS OFF after lag q	Tails off (exponential decay or oscillation)	ACF: count significant spikes → order q
ARMA(p,q)	Tails off after lag q	Tails off after lag p	Both tail off → harder, need AIC/BIC
White Noise	All near zero (within confidence bands)	All near zero	No model needed
Non-Stationary	Slow decay (all lags large)	First PACF ≈ 1.0 (unit root signal)	Difference the series first!

■ Pro Tip

- THE GOLDEN RULE: 'ACF cuts off → MA. PACF cuts off → AR. Both tail off → ARMA.'
- Think of it as: ACF tells you about MA order. PACF tells you about AR order.
- If you can only remember ONE thing: AR uses PACF to find order. MA uses ACF to find order.

10.5 Side-by-Side Comparison: ACF vs PACF

Feature	ACF	PACF
Full name	AutoCorrelation Function	Partial AutoCorrelation Function

What it measures	Total (direct + indirect) correlation of Y_t with Y_{t-k}	DIRECT correlation of Y_t with Y_{t-k} (controlling for lags 1 to $k-1$)
Lag 1 value	$\rho_1 = r_1$	$\phi_{11} = \rho_1$ (same as ACF at lag 1)
Purpose in model ID	Identifies MA order q	Identifies AR order p
MA(q) pattern	Cuts off after lag q ■	Tails off gradually
AR(p) pattern	Tails off gradually	Cuts off after lag p ■
Range	-1 to +1	-1 to +1
Confidence bands	$\pm 1.96/\sqrt{T}$	$\pm 1.96/\sqrt{T}$

■ Interview Questions

- Q: What is the difference between ACF and PACF? [ACF measures total correlation between Y_t and Y_{t-k} including indirect effects. PACF measures the DIRECT correlation between Y_t and Y_{t-k} after removing the effects of intermediate lags.]
- Q: How do you use ACF and PACF to identify AR vs MA models? [ACF cuts off after lag $q \rightarrow$ MA(q). PACF cuts off after lag $p \rightarrow$ AR(p).]
- Q: What does PACF at lag 2 tell you? [It tells you the DIRECT correlation between Y_t and Y_{t-2} , after removing the effect of lag 1.]
- Q: In an AR(2) model, what pattern does the PACF show? [Significant spikes at lags 1 and 2, then sharply cuts off to near zero.]
- Q: Why is PACF at lag 1 equal to ACF at lag 1? [Because at lag 1, there are no intermediate lags to control for — so partial correlation is the same as total correlation.]

■ Memory Tricks

- ACF $\rightarrow$ MA order q (both start with vowels: A and M are first letters, or think: ACF for MoveAhead = MA)
- PACF $\rightarrow$ AR order p (PACF = Partial = AR uses Past directly = AR)
- GOLDEN RULE: 'ACF Cuts Off = MA. PACF Cuts Off = AR.' Memorize this forever.
- AR tail in ACF, AR cut in PACF. MA cut in ACF, MA tail in PACF. Perfect opposites!

CHAPTER 11

White Noise

The simplest stochastic process — and the benchmark for all models

11.1 What is White Noise?

White noise is a sequence of **uncorrelated, identically distributed random variables** with a **constant mean (usually zero)** and **constant variance**. It is the simplest possible stationary time series — there is **no pattern, no correlation, no structure**. It is purely random.

Why 'white' noise? The name comes from physics — white light contains all frequencies equally. Similarly, white noise has equal power at all frequencies. There are no dominant periodicities or patterns.

11.2 Formal Definition and Properties

A process $\{\epsilon_t\}$ is white noise if it satisfies:

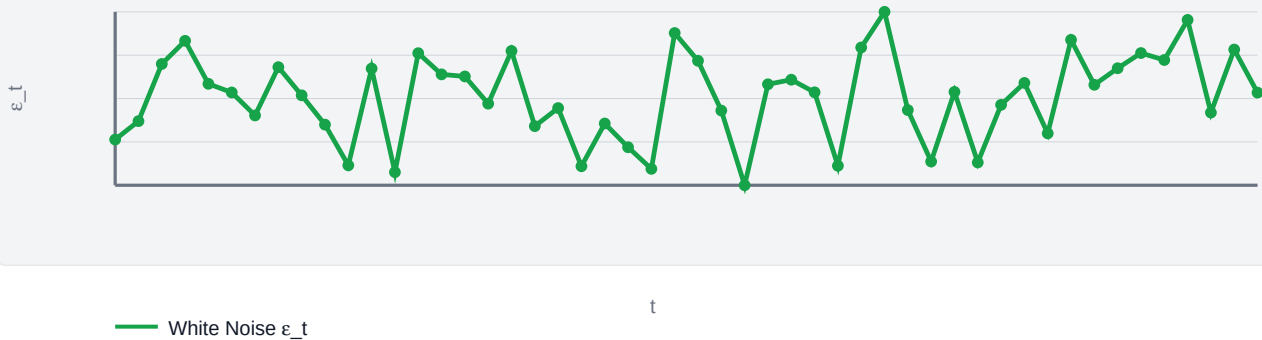
$$E[\epsilon_t] = 0 \text{ (zero mean)}$$

$$\text{Var}(\epsilon_t) = \sigma^2 \text{ (constant, finite variance)}$$

$$\text{Cov}(\epsilon_t, \epsilon_s) = 0 \text{ for all } t \neq s \text{ (zero autocorrelation)}$$

Property	Symbol/Formula	Meaning
Mean	$E[\epsilon_t] = 0$	Fluctuates around zero — no trend
Variance	$\text{Var}(\epsilon_t) = \sigma^2$ (finite)	Constant spread — homoscedastic
Autocovariance	$\gamma_k = 0$ for $k \neq 0$	No correlation between any two time points
ACF	$\rho_k = 0$ for all $k \geq 1$	All ACF values are zero (within noise)
Stationarity	YES — all three conditions satisfied	Mean=0, Var= σ^2 , autocovariance=0
Predictability	NONE	Cannot forecast white noise — it's purely random

Figure 11.1 — White Noise: Random, no pattern, constant variance



White noise: no trend, no seasonality, no autocorrelation. This is the IDEAL residual from a well-fitted model.

11.3 Types of White Noise

Type	Additional Assumption	Notation	Use
Weak White Noise	Only uncorrelated (not necessarily independent)	$\epsilon_t \sim WN(0, \sigma^2)$	Most time series models use this
Strict White Noise	Independently and identically distributed (i.i.d.)	$\epsilon_t \sim i.i.d.(0, \sigma^2)$	Stronger assumption — all moments identical
Gaussian White Noise	i.i.d. + normally distributed	$\epsilon_t \sim N(0, \sigma^2)$	Classical ARIMA residuals — most common

11.4 Why White Noise Matters

White noise is THE benchmark in time series:

- After fitting any model (ARIMA, etc.), the RESIDUALS should be white noise. If they're not, your model has missed some structure.
- A good model = extracts all systematic patterns → leaves only white noise residuals.
- Ljung-Box test checks if residuals are white noise (H_0 : residuals are white noise).
- If original data IS white noise, no time series model can improve forecasting beyond the naive mean forecast.
- White noise is a strictly stationary process — the joint distribution doesn't change with time.

■ Interview Questions

- Q: What is white noise? Define it formally. [White noise $\{\epsilon_t\}$ satisfies: $E[\epsilon_t]=0$, $\text{Var}(\epsilon_t)=\sigma^2$, and $\text{Cov}(\epsilon_t, \epsilon_s)=0$ for $t \neq s$]
- Q: Is white noise stationary? [YES. All three conditions for weak stationarity are satisfied: constant mean (0), constant variance (σ^2), and zero autocorrelation]
- Q: What should the ACF of white noise look like? [All autocorrelation values should be near ZERO and within the $\pm 1.96/\sqrt{T}$ bounds]
- Q: Why do we want model residuals to be white noise? [Because if residuals are white noise, it means the model has captured all systematic patterns]
- Q: What is the difference between weak and Gaussian white noise? [Weak WN: uncorrelated (not necessarily independent); Gaussian WN: i.i.d. + normal distribution]

■ Memory Tricks

- White noise = ZERO pattern. Mean zero, Variance constant, Correlation zero. Three zeros.
- Think of TV static (snow) — completely random. That's white noise.
- After model fitting, CHECK: are residuals white noise? YES → good model. NO → model needs improvement.

CHAPTER 12

Random Walk

The stock price model, unit root, and non-stationarity explained

12.1 What is a Random Walk?

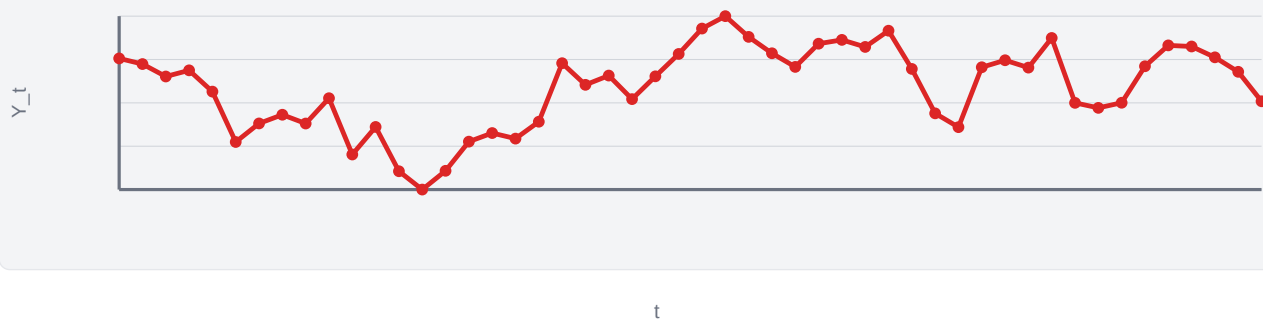
A **random walk** is a process where **each value is the previous value plus a random shock**. Think of a drunk person walking — each step is in a random direction, starting from wherever they ended up on the previous step.

$$\text{Random Walk: } Y_t = Y_{t-1} + \varepsilon_t$$

where $\varepsilon_t \sim \text{WN}(0, \sigma^2)$ is white noise. Starting value Y_0 is a constant.

Key insight: This is exactly an AR(1) with coefficient $\phi = 1$. This value of 1 is called a **unit root**.

Figure 12.1 — Random Walk: No fixed level, wanders unpredictably



Notice: The series **DRIFTS** away from zero — it does **NOT** return to any fixed level. This is **NON-STATIONARY**.

12.2 Properties of Random Walk

Starting from $Y_0 = 0$:

$$Y_t = \varepsilon_t + \varepsilon_{t-1} + \dots + \varepsilon_1 \text{ (cumulative sum of all past shocks)}$$

Property	Formula	Value / Implication
Mean	$E[Y_t]$	= 0 (or Y_0 if non-zero start) — constant ■
Variance	$\text{Var}(Y_t)$	= $t\sigma^2$ — GROWS with time ■ (non-stationary in variance!)
Autocovariance	$\text{Cov}(Y_t, Y_s)$	= $\min(t,s) \cdot \sigma^2$ — depends on t and s, not just lag ■
Autocorrelation	$\text{Corr}(Y_t, Y_s)$	= $\sqrt{\min(t,s)/\max(t,s)}$ — close to 1 for nearby t,s
Stationarity	—	NOT stationary — variance grows with t
Predictability	—	Best forecast: ■ $_{t+1} = Y_t$ (last observation)

12.3 Random Walk WITH Drift

Adding a constant δ (**drift**) gives a random walk with a systematic direction:

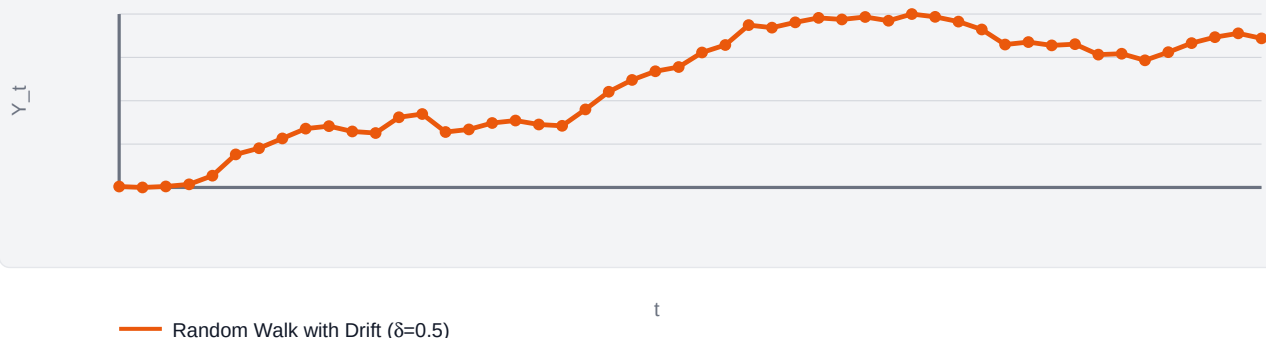
$$\text{Random Walk with Drift: } Y_t = \delta + Y_{t-1} + \varepsilon_t$$

where δ is the drift parameter.

$E[Y_t] = Y_0 + \delta t$ — the mean INCREASES linearly over time (like a trending stock price).

$\text{Var}(Y_t) = t\sigma^2$ — variance still grows with time.

Figure 12.2 — Random Walk WITH Drift: Upward trend on average



Drift adds a systematic upward direction — like a stock market that grows on average but fluctuates randomly.

12.4 Why Stock Prices Are Modelled as Random Walk

The **Efficient Market Hypothesis (EMH)** suggests stock prices follow a random walk. Why? Because:

- All current information is already reflected in the current price.
- New information arrives randomly and unpredictably.
- Therefore, price CHANGES (not levels) are random and unpredictable.
- Best forecast of tomorrow's price = today's price (random walk without drift).
- This is why stock price RETURNS ($\Delta \log(\text{price})$) are often modelled as white noise.

12.5 Random Walk vs Stationary Process

Feature	Random Walk	Stationary AR(1) ($ \phi < 1$)
Formula	$Y_t = Y_{t-1} + \varepsilon_t$ ($\phi=1$)	$Y_t = \phi Y_{t-1} + \varepsilon_t$, $ \phi < 1$
Mean	$E[Y_t] = Y_0$ (constant)	$E[Y_t] = 0$ (constant) ■
Variance	$\text{Var}(Y_t) = t\sigma^2$ (GROWS!) ■	$\text{Var}(Y_t) = \sigma^2/(1-\phi^2)$ (constant) ■
Stationarity	NOT STATIONARY	STATIONARY
ACF	Decays very slowly (close to 1)	Decays geometrically (quickly)
PACF at lag 1	≈ 1.0 (unit root signal)	$= \phi$ (between -1 and +1)

Shock persistence	PERMANENT — shocks never die out	TEMPORARY — shocks decay over time
Forecast	Best = last value Y_t	Returns to mean μ
Fix	Apply first differencing	Model directly as AR(1)

12.6 Fixing a Random Walk — First Differencing

If $Y_t = Y_{t-1} + \varepsilon_t$, then:

$$\Delta Y_t = Y_t - Y_{t-1} = \varepsilon_t \sim \text{WN}(0, \sigma^2)$$

The first difference of a random walk is **white noise** — perfectly stationary! This is why stock price RETURNS (first differences of log prices) are often near white noise.

■ Interview Questions

- Q: What is a random walk? [$Y_t = Y_{t-1} + \varepsilon_t$. Each value equals the previous plus a random shock. It has a unit root (
- Q: Is a random walk stationary? [NO. Although the mean $E[Y_t]=0$ is constant, the variance $\text{Var}(Y_t)=t\sigma^2$ grows with time –
- Q: What is a unit root? [When the AR coefficient $\phi=1$, giving $Y_t=Y_{t-1}+\varepsilon_t$ (random walk). A unit root causes non-stati
- Q: How do you make a random walk stationary? [Apply first differencing: $\Delta Y_t = Y_t - Y_{t-1} = \varepsilon_t$, which is white noise
- Q: What does 'permanent shock' mean in a random walk? [In a random walk, any shock ε_t persists FOREVER — it never
- Q: Why is a random walk a good model for stock prices? [Because the Efficient Market Hypothesis says all information is

■ Tricky / Trap Questions

- TRAP: 'Random walk has a constant mean, so it must be stationary.' WRONG! Stationarity requires constant mean AND c
- TRAP: 'Differencing always makes a series stationary.' TRUE for random walks (I(1) processes). But if the non-stationari
- TRAP: 'AR(1) with $\phi=0.99$ is almost the same as a random walk.' Technically $\phi=0.99 < 1$ so it IS stationary — but it's 'ne

■ Memory Tricks

- Random Walk = Drunk person walking. Each step random. No idea where they'll end up.
- Unit root ($\phi=1$) → Non-stationary. $|\phi|<1$ → Stationary. The condition $|\phi|<1$ is THE stationarity condition for AR(1).
- Fix random walk: DIFFERENCE IT. $Y_t - Y_{t-1} = \text{white noise}$. Simple and clean.

CHAPTER 13

Autoregressive (AR) Models

Predicting from the past — the backbone of time series modelling

13.1 What is an AR Model?

An **Autoregressive (AR) model** predicts the current value of a time series as a **linear combination of its own past values** plus a random error. The word 'auto' means 'self' — it regresses on itself.

Simple Analogy: Your exam score today depends on your last few exam scores. If you did well recently, you'll probably do well again (positive autocorrelation). An AR model captures exactly this kind of 'momentum' or 'memory' in a series.

13.2 AR(1) Model — The Foundation

The **AR(1)** model uses only the **immediately previous value**:

$$\text{AR(1): } Y_t = \phi_1 \cdot Y_{t-1} + \varepsilon_t$$

where ϕ_1 is the AR coefficient and $\varepsilon_t \sim \text{WN}(0, \sigma^2)$.

ϕ_1 value	Behaviour	Stationarity	Example
$0 < \phi_1 < 1$	Gradual positive decay — series slowly returns to mean	STATIONARY ■	Temperature, economic series
$-1 < \phi_1 < 0$	Oscillating negative decay — alternates above/below mean	STATIONARY ■	Some financial ratios
$\phi_1 = 1$	Random walk — variance grows forever	NOT STATIONARY ■	Stock prices
$\phi_1 = -1$	Alternating walk — never settles	NOT STATIONARY ■	Rarely seen
$ \phi_1 > 1$	Explosive — grows without bound	NOT STATIONARY ■	Explosive bubble

Stationarity Condition for AR(1):

$$\text{AR(1) is stationary} \quad \blacksquare \quad |\phi_1| < 1$$

Properties of Stationary AR(1):

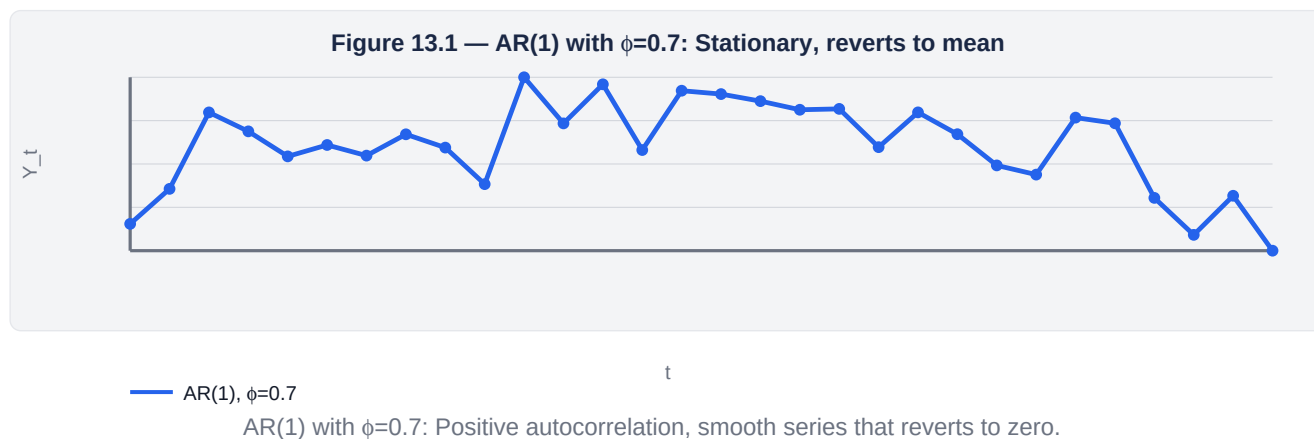
$$\text{Mean: } E[Y_t] = \mu = 0 \text{ (if no constant term)}$$

$$\text{Variance: } \text{Var}(Y_t) = \sigma^2_{\varepsilon} / (1 - \phi_1^2)$$

Autocorrelation: $\rho_k = \phi_1^k$ (geometric decay)

■ **Worked Example — AR(1) Simulation:**

AR(1): $Y_t = 0.7 \cdot Y_{t-1} + \varepsilon_t$, where $\varepsilon_t \sim N(0,1)$. Start $Y_0 = 0$.



13.3 AR(2) Model

The **AR(2)** model uses the last **two** values:

$$\text{AR(2): } Y_t = \phi_1 \cdot Y_{t-1} + \phi_2 \cdot Y_{t-2} + \varepsilon_t$$

Stationarity conditions for AR(2) (all three must hold):

$$\phi_1 + \phi_2 < 1 \quad \text{AND} \quad \phi_2 - \phi_1 < 1 \quad \text{AND} \quad |\phi_2| < 1$$

■ **Worked Example — AR(2) Forecasting:**

Model: $Y_t = 0.5 \cdot Y_{t-1} + 0.3 \cdot Y_{t-2} + \varepsilon_t$

Given: $Y_{100}=10$, $Y_{99}=8$. Forecast Y_{101} :

$$\blacksquare_{101} = 0.5 \times 10 + 0.3 \times 8 = 5.0 + 2.4 = \mathbf{7.4}$$

Forecast Y_{102} : $\blacksquare_{102} = 0.5 \times \blacksquare_{101} + 0.3 \times Y_{100} = 0.5 \times 7.4 + 0.3 \times 10 = 3.7 + 3.0 = \mathbf{6.7}$

13.4 General AR(p) Model

$$\text{AR(p): } Y_t = \phi_1 \cdot Y_{t-1} + \phi_2 \cdot Y_{t-2} + \dots + \phi_p \cdot Y_{t-p} + \varepsilon_t$$

Using **Backshift Operator B**: $B^k Y_t = Y_{t-k}$

$$\text{AR(p) compact: } (1 - \phi_1 \cdot B - \phi_2 \cdot B^2 - \dots - \phi_p \cdot B^p) Y_t = \varepsilon_t$$

$$\Phi(B) Y_t = \varepsilon_t \text{ where } \Phi(B) \text{ is the AR polynomial}$$

13.5 ACF and PACF of AR Models

Model	ACF Pattern	PACF Pattern	Key Identification Rule
AR(1), $\phi > 0$	Positive, exponential decay	Single spike at lag 1, then zero	PACF cuts off after lag 1
AR(1), $\phi < 0$	Alternating +/–, decaying	Single spike at lag 1 (negative)	PACF cuts off after lag 1
AR(2)	Exponential/oscillating decay	Spikes at lags 1 AND 2, then zero	PACF cuts off after lag 2
AR(p)	Exponential/oscillating decay	Spikes at lags 1 to p, then zero	PACF cuts off after lag p

■ Interview Questions

- Q: What is an AR(p) model? [$Y_t = \phi_1 Y_{t-1} + \dots + \phi_p Y_{t-p} + \varepsilon_t$]. It expresses the current value as a linear combination of its own past values and a random error term.
- Q: What is the stationarity condition for AR(1)? [$|\phi_1| < 1$]. If $|\phi_1| \geq 1$, the series is explosive or a unit root (random walk).
- Q: What does the ACF of an AR(1) with $\phi=0.7$ look like? [Exponential decay: $\rho_k = 0.7^k$. So $\rho_1=0.7$, $\rho_2=0.49$, $\rho_3=0.343$].
- Q: What does the PACF of an AR(2) look like? [Significant at lags 1 and 2, then cuts off sharply to near zero. This is the identification rule for AR(2)].
- Q: What is the variance of a stationary AR(1)? [$\text{Var}(Y_t) = \sigma^2_\varepsilon / (1 - \phi_1^2)$]. As $|\phi_1| \rightarrow 1$, variance $\rightarrow \infty$ (consistent with non-stationarity).

■ Memory Tricks

- AR = Autoregressive = 'Self-Regression'. It regresses Y_t on its own past values.
- AR stationarity: |all roots of AR polynomial| > 1 (roots lie OUTSIDE unit circle).
- PACF cuts off after lag p $\rightarrow$ AR(p). This is your model identification rule.

CHAPTER 14

Moving Average (MA) Models

Modelling the impact of past shocks on the current value

14.1 What is an MA Model?

A **Moving Average (MA) model** expresses the current value as a **linear combination of current and past random shocks (errors)**. Do NOT confuse this with the moving average smoothing technique — these are very different!

Intuition: A news announcement (shock) affects today's price and continues to have some effect for q periods. After q periods, the shock's effect completely dies out. An $MA(q)$ model captures exactly this.

14.2 MA(1) Model

$$MA(1): Y_t = \varepsilon_t + \theta_1 \cdot \varepsilon_{t-1}$$

where $\varepsilon_t \sim WN(0, \sigma^2)$ and θ_1 is the MA coefficient.

Important: MA models are **ALWAYS stationary** (regardless of θ value). Why? Because Y_t is a finite linear combination of white noise terms, which are always stationary.

Properties of MA(1):

$$\text{Mean: } E[Y_t] = 0$$

$$\text{Variance: } \text{Var}(Y_t) = \sigma^2(1 + \theta_1^2)$$

$$\text{ACF: } \rho_1 = \theta_1/(1+\theta_1^2), \quad \rho_k = 0 \text{ for } k \geq 2$$

The ACF of MA(1) cuts off to ZERO after lag 1. This is the diagnostic signature!

14.3 MA(q) General Model

$$MA(q): Y_t = \varepsilon_t + \theta_1 \cdot \varepsilon_{t-1} + \theta_2 \cdot \varepsilon_{t-2} + \dots + \theta_q \cdot \varepsilon_{t-q}$$

Using Backshift Operator:

$$Y_t = \Theta(B) \cdot \varepsilon_t \text{ where } \Theta(B) = 1 + \theta_1 B + \theta_2 B^2 + \dots + \theta_q B^q$$

ACF of MA(q):

$$\rho_k = 0 \text{ for all } k > q \text{ (CUTS OFF after lag } q)$$

$$\rho_k \neq 0 \text{ for } k \leq q$$

14.4 Invertibility Condition

Invertibility is to MA models what stationarity is to AR models. An MA(q) model is **invertible** if it can be rewritten as an infinite AR process. This ensures a unique representation and meaningful interpretation.

$$\text{MA(1) is invertible} \blacksquare |\theta_1| < 1$$

Why does invertibility matter? Without it, two different MA models can produce identical ACF patterns — making parameter estimation non-unique. Invertibility ensures the model has a unique, sensible interpretation.

14.5 MA(1) Worked Example

Model: $Y_t = \varepsilon_t + 0.6 \cdot \varepsilon_{t-1}$. Given shocks: $\varepsilon_1=1.5$, $\varepsilon_2=-0.8$, $\varepsilon_3=1.2$, $\varepsilon_4=-0.5$

t	ε_t (shock)	$\theta_1 \cdot \varepsilon_{t-1}$	$Y_t = \varepsilon_t + \theta_1 \cdot \varepsilon_{t-1}$
1	1.5	0 (no past shock)	$Y_1 = 1.5 + 0 = 1.5$
2	-0.8	$0.6 \times 1.5 = 0.9$	$Y_2 = -0.8 + 0.9 = 0.1$
3	1.2	$0.6 \times (-0.8) = -0.48$	$Y_3 = 1.2 - 0.48 = 0.72$
4	-0.5	$0.6 \times 1.2 = 0.72$	$Y_4 = -0.5 + 0.72 = 0.22$
5 (Forecast)	0 (expected shock=0)	$0.6 \times (-0.5) = -0.30$	$\blacksquare_5 = 0 + (-0.30) = -0.30$

Interpretation: The forecast for period 5 is -0.30, reflecting the lingering effect of the last shock $\varepsilon_4=-0.5$.

14.6 Comparing AR and MA Models

Feature	AR(p) Model	MA(q) Model
What drives Y_t	Past VALUES of Y	Past ERRORS (shocks)
Stationarity	Required: $ \text{roots} > 1$ outside unit circle	ALWAYS stationary
Invertibility	ALWAYS invertible	Required: $ \text{roots} > 1$ outside unit circle

Memory	Infinite (decaying) — all past values have some influence	Finite — only last q shocks matter
ACF	Tails off (exponential/oscillating decay)	Cuts off after lag q
PACF	Cuts off after lag p	Tails off (exponential/oscillating decay)
Parameters	$\phi_1, \dots, \phi_p$	$\theta_1, \dots, \theta_q$
Identification	Look at PACF → count significant lags	Look at ACF → count significant lags

■ Interview Questions

- Q: What is an MA(q) model? [$Y_t = \varepsilon_t + \theta_1\varepsilon_{t-1} + \dots + \theta_q\varepsilon_{t-q}$]. It expresses Y_t as a linear combination of the current and past errors.
- Q: Are MA models always stationary? [YES! An MA(q) process is always stationary regardless of the θ parameters. It's a finite moving average of a stationary process.]
- Q: What is invertibility in an MA model? [An MA model is invertible if it can be rewritten as an infinite AR process. Ensure the roots of the MA polynomial are outside the unit circle.]
- Q: What does the ACF of MA(2) look like? [Significant spikes at lags 1 and 2, then ZERO for all lags ≥ 3 . The sharp cutoff after lag 2 is characteristic.]
- Q: What is the key difference between AR and MA models? [AR: Y_t depends on past LEVELS. MA: Y_t depends on past ERRORS.]

■ Memory Tricks

- MA = Moving Average of ERRORS (not of Y !). Huge common confusion.
- MA ACF CUTS OFF — like scissors cutting. AR ACF TAILS OFF — like a tail of a cat.
- MA always stationary. AR needs $|\phi| < 1$. Only MA needs invertibility condition.

CHAPTER 15

ARMA Models

Combining the strengths of AR and MA in one powerful model

15.1 Why ARMA? Combining AR and MA

In practice, many time series cannot be adequately described by a pure AR(p) or pure MA(q) model. The **ARMA(p,q) model** combines both components, providing a **more parsimonious** (fewer parameters) representation of complex autocorrelation structures.

Key insight: An AR model with many lags might need $p=10+$ parameters to fit some data. An ARMA model might capture the same structure with $p=1, q=1$ — just 2 parameters. This is the principle of **parsimony** — fewest parameters that adequately fit the data.

15.2 ARMA(p,q) Formula

$$\text{ARMA}(p,q): Y_t = \phi_1 Y_{t-1} + \dots + \phi_p Y_{t-p} + \varepsilon_t + \theta_1 \varepsilon_{t-1} + \dots + \theta_q \varepsilon_{t-q}$$

In compact backshift notation:

$$\Phi(B) \cdot Y_t = \Theta(B) \cdot \varepsilon_t$$

where $\Phi(B) = 1 - \phi_1 B - \dots - \phi_p B^p$ (AR polynomial)

and $\Theta(B) = 1 + \theta_1 B + \dots + \theta_q B^q$ (MA polynomial)

15.3 Special Cases of ARMA

Model	ARMA form	Interpretation
AR(p)	ARMA(p, 0)	Pure autoregressive — no MA component
MA(q)	ARMA(0, q)	Pure moving average — no AR component
ARMA(1,1)	ARMA(1, 1)	AR(1) + MA(1): $Y_t = \phi_1 Y_{t-1} + \varepsilon_t + \theta_1 \varepsilon_{t-1}$
White Noise	ARMA(0, 0)	No AR, no MA — just ε_t

15.4 ARMA(1,1) — Most Common in Practice

$$\text{ARMA}(1,1): Y_t = \phi_1 \cdot Y_{t-1} + \varepsilon_t + \theta_1 \cdot \varepsilon_{t-1}$$

Conditions:

- Stationarity: $|\phi_1| < 1$ (same as AR(1) condition)
- Invertibility: $|\theta_1| < 1$ (same as MA(1) condition)

Properties of ARMA(1,1):

$$\text{Mean: } E[Y_t] = 0$$

$$\text{Variance: } \text{Var}(Y_t) = \sigma^2(1 + 2\phi_1\theta_1 + \theta_1^2) / (1 - \phi_1^2)$$

$$\rho_1 = (\phi_1 + \theta_1)(1 + \phi_1\theta_1) / (1 + 2\phi_1\theta_1 + \theta_1^2)$$

$$\rho_k = \phi_1^{k-1} \cdot \rho_1 \text{ for } k \geq 2 \text{ (geometric decay after lag 1)}$$

15.5 ACF and PACF of ARMA — The Tricky Pattern

Here's the key challenge with ARMA: **BOTH the ACF and PACF tail off**. There is no sharp cutoff to identify the order easily.

Model	ACF Pattern	PACF Pattern	Identification method
AR(p)	Tails off gradually	Cuts off after lag p	Easy: count PACF spikes
MA(q)	Cuts off after lag q	Tails off gradually	Easy: count ACF spikes
ARMA(p,q)	Tails off after lag (q-p)	Tails off after lag (p-q)	Hard! Use AIC/BIC to select p and q

■ Pro Tip

- For ARMA model selection, use INFORMATION CRITERIA: $AIC = -2\ln(L) + 2k$, $BIC = -2\ln(L) + k \cdot \ln(T)$. Choose model with lowest value.
- In practice, try multiple (p,q) combinations and select the best using AIC/BIC.

15.6 Model Selection Criteria

Criterion	Formula	Penalty for	When to prefer
AIC (Akaike)	$-2 \cdot \ln(L) + 2k$	Number of parameters (mild)	Larger samples, when prediction is goal
BIC (Bayesian)	$-2 \cdot \ln(L) + k \cdot \ln(T)$	Number of parameters (heavy)	Smaller samples, when model truth is goal
AICc	$AIC + 2k(k+1)/(T-k-1)$	Corrected for small samples	When T is small relative to k

■ Interview Questions

- Q: What is an ARMA(p,q) model? [It combines AR(p) and MA(q): $Y_t = \phi_1 Y_{t-1} + \dots + \phi_p Y_{t-p} + \varepsilon_t + \theta_1 \varepsilon_{t-1} + \dots + \theta_q \varepsilon_{t-q}$]
- Q: When would you use ARMA instead of pure AR or MA? [When both ACF and PACF tail off — suggesting neither pure AR nor MA]
- Q: What are the conditions for ARMA(1,1) to be stationary and invertible? [Stationary: $|\phi_1| < 1$. Invertible: $|\theta_1| < 1$. Both]
- Q: How do you select p and q for an ARMA model? [When both ACF and PACF tail off, try multiple (p,q) combinations and compare model fits]

CHAPTER 16

ARIMA Models ■■ MOST IMPORTANT CHAPTER

The complete framework — AR + Integration + MA for real-world series

■■ Why ARIMA is the Centrepiece

- ARIMA is the most widely used time series model in practice.
- It handles NON-STATIONARY data by incorporating differencing.
- ARIMA interview questions are guaranteed in ANY MSc Statistics interview.
- Understanding ARIMA completely = understanding time series modelling.
- Box-Jenkins methodology for ARIMA is the standard 4-step approach examiners love.

16.1 What Does ARIMA Stand For?

Letter	Full Name	What it does	Parameter
AR	AutoRegressive	Models dependence on own past values	p = AR order
I	Integrated	Handles non-stationarity via differencing	d = differencing order
MA	Moving Average	Models dependence on past forecast errors	q = MA order

ARIMA(p, d, q) — Three parameters: p (AR order), d (differencing), q (MA order)

16.2 The ARIMA Formula — Step by Step

ARIMA(p, d, q) works in two stages:

Stage 1: Difference the series d times to make it stationary:

$$W_t = \Delta^d Y_t \text{ (d-th difference of } Y_t\text{)}$$

Stage 2: Fit an ARMA(p, q) model to the differenced series W_t :

$$\Phi(B) \cdot W_t = \Theta(B) \cdot \varepsilon_t$$

Combined in one equation:

$$\Phi(B) \cdot (1-B)^d \cdot Y_t = \Theta(B) \cdot \varepsilon_t$$

Important Special Cases:

ARIMA notation	Equivalent model	Description
ARIMA(0,0,0)	White noise	$Y_t = \varepsilon_t$ — purely random
ARIMA(1,0,0)	AR(1)	Stationary AR — no differencing needed
ARIMA(0,0,1)	MA(1)	Stationary MA — no differencing needed
ARIMA(1,0,1)	ARMA(1,1)	Stationary ARMA
ARIMA(0,1,0)	Random Walk	$Y_t = Y_{t-1} + \varepsilon_t$ — diff once → white noise
ARIMA(1,1,0)	Differenced AR(1)	Most common! Trending series with AR structure
ARIMA(0,1,1)	Differenced MA(1)	Equivalent to Simple Exponential Smoothing!
ARIMA(0,2,2)	—	Equivalent to Holt's Linear Trend method
ARIMA(p,1,q)	—	Typical for financial/economic series with trend

16.3 The Box-Jenkins Methodology — 4 Steps

Box and Jenkins (1970) developed a systematic 4-step approach to ARIMA modelling. This is **THE standard framework** that every interviewer expects you to know.

STEP 1: Identification

- Plot the data — check for trend, seasonality, outliers.
- Check stationarity: rolling mean/variance, ADF test.
- If non-stationary: difference until stationary. Count differences → this is d .
- Plot ACF and PACF of the stationary (differenced) series.
- Identify p and q : PACF cuts off at p → AR(p); ACF cuts off at q → MA(q).

STEP 2: Estimation

- Estimate parameters ($\phi_1, \dots, \phi_p, \theta_1, \dots, \theta_q$) using Maximum Likelihood Estimation (MLE) or least squares.
- Most software (R, Python) does this automatically.
- Check that all estimated parameters are statistically significant (t-test, p-value).

STEP 3: Diagnostic Checking

- Examine residuals — they should be WHITE NOISE.
- Plot ACF of residuals — all should be within confidence bands.
- Run Ljung-Box test (H_0 : residuals are white noise). Want p-value > 0.05 .
- Check residuals for normality (Q-Q plot, Shapiro-Wilk test).
- If residuals are NOT white noise → model is inadequate → go back to Step 1.

STEP 4: Forecasting

- Use the fitted model to generate forecasts.
- Compute forecast intervals (prediction intervals).
- Evaluate forecast accuracy on hold-out data: MAE, RMSE, MAPE.
- If accuracy is poor → revisit model specification.

16.4 Detailed Worked Example — Full ARIMA Process

Problem: Monthly electricity consumption (units) for 2 years. Data shows clear upward trend.

Step 1 — Identification:

- Plot shows upward trend → NOT stationary → apply first differencing ($d=1$).
- After 1st differencing: ADF test p-value = 0.02 < 0.05 → reject H_0 → series is NOW stationary.
- ACF of differenced series: large spike at lag 1, then cuts off → suggests MA(1) → $q=1$.
- PACF of differenced series: exponential decay from lag 1 → tails off → consistent with MA(1).
- Tentative model: ARIMA(0,1,1).

Step 2 — Estimation:

MLE gives: $\theta_1 = 0.65$ (t-statistic = 4.2, $p < 0.001$) — significant.

Step 3 — Diagnostics:

ACF of residuals: all within ± 0.2 bounds. Ljung-Box test: p-value = 0.43 > 0.05 → residuals are white noise ■

Step 4 — Forecasting:

Given $Y_{24} = 520$, $\varepsilon_{24} = 15$: Forecast = ■₂₅ = $Y_{24} + \theta_1 \cdot \varepsilon_{24} = 520 + 0.65 \times 15 = 529.75$

16.5 Understanding p, d, q — Quick Reference

Parameter	Name	What it controls	How to find it	Typical values
p	AR order	How many past Y values to include	Count where PACF cuts off	0, 1, 2 (rarely 3+)
d	Differencing	How many times to difference	Count differences until stationary (ADF test)	0, 1 (rarely 2)
q	MA order	How many past errors to include	Count where ACF cuts off	0, 1, 2 (rarely 3+)

16.6 ARIMA Forecasting Formulas**ARIMA(1,1,1) Example — One Step Ahead Forecast:**

Model on differenced series $W_t = \Delta Y_t$: $W_t = \phi_1 W_{t-1} + \varepsilon_t + \theta_1 \varepsilon_{t-1}$

Forecast: ■_{t+1} = $\phi_1 \cdot W_t + \theta_1 \cdot \varepsilon_t$

Convert back: ■_{t+1} = $Y_t + \text{■}_{t+1}$

■ Numerical Example:

ARIMA(1,1,1): $\phi_1=0.4$, $\theta_1=0.6$. $Y_{\{100\}}=200$, $W_{\{100\}}=\Delta Y_{\{100\}}=5$, $\varepsilon_{\{100\}}=2$.

$$\blacksquare_{\{101\}} = 0.4 \times 5 + 0.6 \times 2 = 2.0 + 1.2 = 3.2$$

$$\blacksquare_{\{101\}} = Y_{\{100\}} + \blacksquare_{\{101\}} = 200 + 3.2 = \mathbf{203.2}$$

■ Interview Questions

- Q: What does ARIMA stand for and what does each part do? [AutoRegressive Integrated Moving Average. AR: models dependence on its own past values. I: differencing to make stationary. MA: models the error term.]
- Q: What is the meaning of ARIMA(2,1,1)? [AR order 2 (depends on $Y_{\{t-1\}}$ and $Y_{\{t-2\}}$), differenced once (to remove trend), MA order 1.]
- Q: What are the 4 steps of Box-Jenkins methodology? [1) Identification (plot, ACF/PACF, choose p,d,q). 2) Estimation (ML or OLS). 3) Diagnostic checking (residuals). 4) Model validation.]
- Q: Is ARIMA(0,1,0) equivalent to what? [It's a Random Walk: $Y_t = Y_{\{t-1\}} + \varepsilon_t$. Differencing once gives white noise, but the series itself is non-stationary.]
- Q: Is ARIMA(0,1,1) equivalent to what? [It is equivalent to Simple Exponential Smoothing! This is a famous and frequent connection.]
- Q: How do you choose d in ARIMA? [Plot the series and check for trend. Compute ADF test. If non-stationary, difference once. If still non-stationary, difference twice.]
- Q: What should the residuals of a well-fitted ARIMA model look like? [They should be white noise: zero mean, constant variance, no autocorrelation.]

■ Tricky / Trap Questions

- TRAP: 'ARIMA always needs differencing (d>0).' WRONG — if the series is already stationary, d=0 and it reduces to ARMA.
- TRAP: 'ARIMA can handle seasonality.' Standard ARIMA CANNOT — you need SARIMA for seasonal data.
- TRAP: 'I can identify p and q from a non-stationary series.' WRONG — always difference FIRST, THEN look at ACF/PACF.
- TRAP: 'Higher p and q always gives better models.' WRONG — more parameters = risk of overfitting. Use AIC/BIC to penalize complexity.

■ Memory Tricks

- ARIMA(p,d,q): p = AR lags, d = differences, q = MA lags. Remember the ORDER: A-R-I-M-A.
- d=1: Linear trend. d=2: Quadratic trend. d=0: Already stationary.
- Box-Jenkins steps: I-E-D-F = Identify → Estimate → Diagnose → Forecast.
- ARIMA(0,1,1) = SES. ARIMA(0,2,2) = Holt's. Important connections!

CHAPTER 17

Seasonal ARIMA (SARIMA)

Extending ARIMA to handle seasonal patterns

17.1 Why SARIMA?

Standard ARIMA cannot handle seasonality. When data has a repeating seasonal pattern (e.g., monthly, quarterly), we use **SARIMA** — **Seasonal ARIMA**. SARIMA adds **seasonal counterparts** to each of the three ARIMA components.

17.2 SARIMA Notation — Understanding (p,d,q)(P,D,Q)[s]

$$\text{SARIMA}(p, d, q)(P, D, Q)[s]$$

Parameter	Type	Meaning	Example
p	Non-seasonal	Non-seasonal AR order	AR at lag 1,2,...,p
d	Non-seasonal	Non-seasonal differencing order	Remove trend
q	Non-seasonal	Non-seasonal MA order	MA at lag 1,2,...,q
P	Seasonal	Seasonal AR order	AR at lags s, 2s,...,Ps
D	Seasonal	Seasonal differencing order	Remove seasonality
Q	Seasonal	Seasonal MA order	MA at lags s, 2s,...,Qs
s	Period	Length of seasonal cycle	s=12 monthly, s=4 quarterly, s=7 daily-weekly

17.3 SARIMA Full Formula

The SARIMA model combines non-seasonal and seasonal polynomials:

$$\Phi(B) \cdot \Phi_s(B^s) \cdot (1-B)^d \cdot (1-B^s)^D \cdot Y_t = \Theta(B) \cdot \Theta_s(B^s) \cdot \varepsilon_t$$

where:

- $\Phi(B)$ = non-seasonal AR polynomial (order p)
- $\Phi_s(B^s)$ = seasonal AR polynomial (order P, at multiples of s)
- $(1-B)^d$ = non-seasonal differencing
- $(1-B^s)^D$ = seasonal differencing
- $\Theta(B)$ = non-seasonal MA polynomial (order q)
- $\Theta_s(B^s)$ = seasonal MA polynomial (order Q, at multiples of s)

17.4 Worked Example — SARIMA(1,1,1)(1,1,1)[12]

This is a common model for monthly data with trend and seasonality (like airline passengers).

Data: Monthly airline passenger numbers. Trend: Yes (growing). Seasonality: Yes (annual, $s=12$). Variance: Increasing → take log first.

Steps:

- Take $\log(Y_t)$ to stabilise variance (makes multiplicative → additive).
- Apply seasonal differencing ($D=1, s=12$): remove the annual seasonal pattern.
- Apply non-seasonal differencing ($d=1$): remove remaining trend.
- Check ACF/PACF of twice-differenced log series.
- Identify non-seasonal p, q (from ACF/PACF at non-seasonal lags) → $p=1, q=1$.
- Identify seasonal P, Q (from ACF/PACF at lags 12, 24, ...) → $P=1, Q=1$.
- Final model: SARIMA(1,1,1)(1,1,1)[12] on $\log(Y_t)$.

17.5 ACF/PACF Patterns for Seasonal Models

Seasonal patterns appear at **multiples of s** in the ACF/PACF:

Pattern in ACF/PACF	Indicates	Model suggestion
Large spike at lag s (and maybe $2s, 3s$) in ACF, cuts off	Seasonal MA component	Add Q (seasonal MA order)
Gradual decay at lags $s, 2s, 3s$ in ACF	Seasonal AR component	Add P (seasonal AR order)
Large spike at lag s in PACF, cuts off	Seasonal AR(1) component	$P=1$
Significant spikes at lags 1 AND 12 in ACF	Both regular + seasonal MA	SARIMA with $q=1, Q=1$

17.6 Choosing D and s

Data frequency	Seasonal period s	$D=1$ removes
Monthly	12	Annual seasonality
Quarterly	4	Annual seasonality
Weekly	52	Annual seasonality
Daily	7	Weekly seasonality
Daily	365	Annual seasonality
Hourly	24	Daily seasonality

■ Pro Tip

- In practice, $D=0$ or $D=1$ is almost always sufficient. $D=2$ is very rarely needed.
- Apply seasonal differencing BEFORE non-seasonal differencing. Check stationarity after each step.
- Famous SARIMA example: airline passengers data → $\text{SARIMA}(0,1,1)(0,1,1)[12]$ — this is the 'airline model' by Box and Jenkins

■ Interview Questions

- Q: What is SARIMA and how does it extend ARIMA? [SARIMA = Seasonal ARIMA. It adds seasonal $\text{AR}(P)$, seasonal differencing, and seasonal moving average.]
- Q: What does $\text{SARIMA}(0,1,1)(0,1,1)[12]$ represent? [Non-seasonal: $d=1$ (one regular difference), $q=1$ (MA at lag 1). Seasonal: $P=1$ (seasonal AR at lag 12), $Q=1$ (seasonal MA at lag 12).]
- Q: How do you identify seasonal parameters P and Q ? [Look at ACF/PACF at seasonal lags ($s, 2s, 3s, \dots$). If ACF cuts off at lag s , it suggests $\text{AR}(P)$. If it decays slowly, it suggests $\text{MA}(Q)$.]
- Q: Why would you take logs before fitting SARIMA? [To stabilise variance when the seasonal amplitude increases proportionally with the level of the series.]

CHAPTER 18

Model Identification

The complete step-by-step ACF/PACF decision framework

18.1 The Complete Identification Framework

Model identification is the **most difficult and most tested** part of ARIMA modelling. Here is a **systematic, foolproof framework** to follow in any interview or exam.

Step 1: Plot the raw data

- Look for: trend, seasonality, changing variance, outliers.
- If variance increases: apply log or square-root transformation.
- If seasonal pattern present: note the period s .

Step 2: Check and achieve stationarity

- Visual: are rolling mean and variance roughly constant?
- Formal: run ADF test (H_0 : non-stationary). If $p > 0.05$, series is non-stationary.
- If non-stationary: apply first differencing. Increment d by 1.
- Repeat ADF test on differenced series. Stop when stationary.

Step 3: Handle seasonality

- If seasonal pattern remains after non-seasonal differencing: apply seasonal differencing.
- For monthly data: $\Delta_{12} Y_t = Y_t - Y_{t-12}$. Increment D by 1.
- Check ACF/PACF for remaining seasonal pattern at lags s , $2s$, $3s$.

Step 4: Read the ACF and PACF

- Plot ACF and PACF of the final stationary series.
- Identify non-seasonal p and q : use golden rule (ACF cutoff $\rightarrow q$, PACF cutoff $\rightarrow p$).
- Identify seasonal P and Q : look at lags s , $2s$ (cutoff $\rightarrow Q$, or P).
- When both tail off: try ARMA $\rightarrow$ use AIC/BIC to select best model.

Step 5: Fit model and diagnose

- Estimate parameters using MLE or conditional least squares.
- Check residuals: ACF should show no significant lags.
- Ljung-Box test: want p-value > 0.05 (cannot reject white noise).
- If residuals show pattern: go back to Step 4, revise p or q.

18.2 ACF/PACF Decision Table — The Master Reference

ACF of stationary series	PACF of stationary series	Model to fit	Notes
Cuts off after lag q	Tails off (decays geometrically)	MA(q)	q = lag where ACF cuts off
Tails off (decays geometrically)	Cuts off after lag p	AR(p)	p = lag where PACF cuts off
Tails off (decays geometrically)	Tails off (decays geometrically)	ARMA(p,q)	Use AIC/BIC to select p and q
All within confidence bands	All within confidence bands	White noise / ARIMA(0,0,0)	No model needed
Slow decay, all lags large	1st lag ≈ 1.0 , then drops	Non-stationary (unit root)	Difference the series — DON'T fit yet
Significant at lags s, 2s, 3s...	Decays at seasonal lags	Seasonal MA component	Include Q in SARIMA
Decays at seasonal lags	Significant at lag s, cuts off	Seasonal AR component	Include P in SARIMA

18.3 Practical Examples of Model Identification

Example 1

Observed: ACF: large spike at lag 1 only, then within bands. PACF: decays geometrically (lags 1,2,3,... getting smaller).

Conclusion: Model: MA(1). The ACF cuts off after lag 1, PACF tails off. Order $q=1$.

Example 2

Observed: ACF: exponential decay from lag 1 (0.7, 0.49, 0.34, ...). PACF: large spike at lag 1, all others within bands.

Conclusion: Model: AR(1) with $\phi=0.7$. PACF cuts off after lag 1, ACF tails off.

Example 3

Observed: ACF: all within confidence bands. PACF: all within bands.

Conclusion: White noise! No model needed — the series cannot be forecast better than its mean.

Example 4

Observed: After first differencing: ACF has spikes at lags 1 and 12. PACF decays at lags 1, 2 and has spike at lag 12.

Conclusion: SARIMA with $d=1$, $q=1$, $Q=1$, $s=12 \rightarrow \text{SARIMA}(0,1,1)(0,0,1)[12]$. Monthly seasonal data.

Example 5

Observed: Both ACF and PACF tail off (but significant at lags 1-3).

Conclusion: ARMA → try combinations. ARMA(1,1), ARMA(2,1), ARMA(1,2). Compare AIC. Choose lowest.

18.4 Useful Rules of Thumb

Rule	Details
Never fit ARIMA to non-stationary data	ALWAYS check and achieve stationarity FIRST before reading ACF/PACF.
Start simple	Try ARIMA(0,1,1) or ARIMA(1,1,0) before complex models. Often sufficient.
d=1 is usually enough	Most economic/financial series are I(1). Rarely need d=2.
Keep $p+q \leq 5$	High-order models overfit. If you need $p=5+$, check if you missed differencing.
Both ACF and PACF tail off → ARMA	Use AIC/BIC to choose, don't guess. Grid search over p and q.
Ljung-Box is the final verdict	If test fails ($p < 0.05$), residuals are NOT white noise → model is inadequate.

■ Interview Questions

- Q: Walk me through how you would identify an ARIMA model for a dataset. [1] Plot data — check for trend, seasonality. 2)
- Q: You see an ACF that decays to zero very slowly. What does this tell you? [The series is NON-STATIONARY. Do NOT a
- Q: Both ACF and PACF are significant at lag 1 only and then drop to zero. What model? [Could be ARMA(1,1), or possibly
- Q: What is the Ljung-Box test and how do you use it? [It tests H_0 : first m autocorrelations of residuals are all zero (w

CHAPTER 19

Forecasting

Predicting the future: one-step, multi-step, and prediction intervals

19.1 What is Forecasting?

Forecasting is the process of using a fitted time series model to **predict future values**. It is the ultimate goal of most time series analyses. A good forecast tells you not just the expected value, but also **how uncertain** that prediction is.

Key terminology:

Term	Meaning
Forecast horizon (h)	How many periods ahead you predict (h=1: one-step, h=12: twelve-step)
Point forecast ($\hat{Y}_{t+h}$)	Single best estimate for Y_{t+h}
Prediction interval	Range containing the true future value with specified probability (e.g., 95%)
Forecast error	$e_t = Y_t - \hat{Y}_t$ (actual minus predicted, known AFTER the fact)
Training set	Historical data used to fit the model
Test set / Hold-out	Recent data WITHHELD to evaluate forecast accuracy

19.2 One-Step Ahead Forecast

A **one-step forecast** predicts one period into the future. This is the simplest case.

For ARIMA(p,d,q): At time T, the one-step forecast:

$$\hat{Y}_{T+1|T} = E[Y_{T+1} | Y_T, Y_{T-1}, \dots, Y_1]$$

Examples:

- **AR(1):** $Y_t = \phi \cdot Y_{t-1} + \varepsilon_t \rightarrow \hat{Y}_{T+1} = \phi \cdot Y_T$
- **MA(1):** $Y_t = \varepsilon_t + \theta \cdot \varepsilon_{t-1} \rightarrow \hat{Y}_{T+1} = \theta \cdot \varepsilon_T$
- **ARMA(1,1):** $Y_t = \phi Y_{t-1} + \varepsilon_t + \theta \varepsilon_{t-1} \rightarrow \hat{Y}_{T+1} = \phi \cdot Y_T + \theta \cdot \varepsilon_T$
- **Random Walk:** $Y_t = Y_{t-1} + \varepsilon_t \rightarrow \hat{Y}_{T+1} = Y_T$ (naive forecast)

19.3 Multi-Step Forecasting

For **h-step ahead forecasts** ($h > 1$), we use the **chain rule of forecasting**: substitute forecasts for unknown future values.

■ AR(1) Multi-Step Forecast:

Model: $Y_t = 0.6 \cdot Y_{t-1} + \varepsilon_t$. Current value: $Y_T = 10$.

Forecast horizon h	Formula	Calculation	$\hat{Y}_{T+h}$
$h = 1$	$\hat{Y}_{T+1} = \phi \cdot Y_T$	0.6×10	6.0
$h = 2$	$\hat{Y}_{T+2} = \phi \cdot \hat{Y}_{T+1}$	0.6×6.0	3.6
$h = 3$	$\hat{Y}_{T+3} = \phi \cdot \hat{Y}_{T+2}$	0.6×3.6	2.16
$h = 4$	$\hat{Y}_{T+4} = \phi \cdot \hat{Y}_{T+3}$	0.6×2.16	1.30
$h \rightarrow \infty$	$\hat{Y}_{T+h} = \phi^h \cdot Y_T \rightarrow 0$	$0.6^h \times 10 \rightarrow 0$	Reverts to mean (0)

Key insight: As h increases, AR(1) forecasts converge to the long-run mean. This is called **mean reversion**.

19.4 Prediction Intervals

A **prediction interval** gives a range within which the actual future value is expected to fall with a specified probability (e.g., 95%). Unlike confidence intervals for parameters, prediction intervals are for **individual future observations**.

$$\text{95\% Prediction Interval: } \hat{Y}_{T+h} \pm 1.96 \times \sigma_h$$

where σ_h is the forecast standard error at horizon h .

For AR(1):

$$\sigma^2_1 = \sigma^2_\varepsilon \text{ (one-step)}$$

$$\sigma^2_h = \sigma^2_\varepsilon \cdot (1 + \phi^2 + \phi^4 + \dots + \phi^{2(h-1)}) \text{ (h-step)}$$

Key property: Prediction intervals **WIDEN** as h increases — the further ahead you forecast, the more uncertain.

■ Pro Tip

- Prediction intervals vs Confidence intervals: CI is for a **PARAMETER** (fixed unknown). PI is for a **FUTURE OBSERVATION**.
- For a random walk: $\sigma^2_h = h \cdot \sigma^2_\varepsilon$ — prediction intervals grow proportionally to $\sqrt{h}$. Very wide for long horizons.
- The width of prediction intervals tells you **HOW UNCERTAIN** your forecasts are. Wide intervals = low predictability.

■ Interview Questions

- Q: What is the difference between a point forecast and a prediction interval? [Point forecast: single best estimate. Pre
- Q: What happens to prediction intervals as the forecast horizon increases? [They WIDEN — uncertainty grows as you fore
- Q: For AR(1) with $\phi=0.8$, $Y_T=10$, what is the 3-step ahead forecast? [$\hat{Y}_{T+1}=0.8 \times 10=8$, $\hat{Y}_{T+2}=0.8 \times 8=6.4$, $\hat{Y}_{T+3}=$
- Q: Why do AR model forecasts eventually revert to the mean? [Because the AR(p) process is stationary with a finite mean

CHAPTER 20

Forecast Accuracy Metrics

MAE, MSE, RMSE, MAPE — measuring how good your forecasts are

20.1 Why Measure Forecast Accuracy?

Before deploying any forecast, you need to know **how accurate it is**. Accuracy metrics help you: compare models, identify whether a model is improving, and communicate forecast quality to stakeholders. All metrics are computed on the **test (hold-out) set** — data the model has NOT seen.

Let $e_t = Y_t - \hat{Y}_t$ be the forecast error at time t .

20.2 MAE — Mean Absolute Error

$$\text{MAE} = (1/n) \cdot \sum |Y_t - \hat{Y}_t| = (1/n) \cdot \sum |e_t|$$

Interpretation: Average absolute difference between actual and predicted. In the SAME units as Y_t .

Example: MAE = 5 units means on average, predictions are 5 units away from actual values.

Feature	MAE
Interpretation	Average absolute error in original units
Sensitivity to outliers	LESS sensitive (uses absolute value, not squared)
Units	Same as the data
Range	0 to ∞ (0 is perfect)
Best for	When you want to penalise all errors equally (no over-penalty for large errors)
Limitation	Doesn't distinguish between over- and under-forecasting

20.3 MSE — Mean Squared Error

$$\text{MSE} = (1/n) \cdot \sum (Y_t - \hat{Y}_t)^2 = (1/n) \cdot \sum e_t^2$$

Key property: SQUARES the errors, so large errors are **penalised much more heavily**.

Units: Squared units (e.g., if Y is in kg, MSE is in kg^2) — hard to interpret directly.

20.4 RMSE — Root Mean Squared Error

$$\text{RMSE} = \sqrt{\text{MSE}} = \sqrt{[(1/n) \cdot \sum (Y_t - \hat{Y}_t)^2]}$$

RMSE fixes the units problem of MSE by taking the square root. Result is in the same units as Y_t .

Comparison: $\text{RMSE} \geq \text{MAE}$ always. If $\text{RMSE} \gg \text{MAE}$, the model has some very large errors (outliers in residuals).

20.5 MAPE — Mean Absolute Percentage Error

$$\text{MAPE} = (100/n) \cdot \sum |Y_t - \hat{Y}_t| / |Y_t|$$

Interpretation: Average error as a PERCENTAGE of the actual value. Dimensionless — can compare across different series!

Example: $\text{MAPE} = 8\%$ means on average, predictions are 8% away from actual values.

Limitation: MAPE is **undefined when $Y_t = 0$** (division by zero). Also asymmetric — it penalises positive errors more than negative ones.

20.6 Summary Comparison Table

Metric	Formula	Units	Outlier sensitive	Best used when
MAE	$(1/n)\sum e_t $	Same as data	Low	Errors are equally important; want robustness
MSE	$(1/n)\sum e_t^2$	Squared units	HIGH	Large errors are especially costly
RMSE	$\sqrt{\text{MSE}}$	Same as data	HIGH	Standard error metric; interpretable units
MAPE	$(100/n)\sum e_t/Y_t $	Percentage (%)	Moderate	Comparing forecast accuracy across different series/s

■ Worked Example — Computing All Metrics:

t	Y_t	$\hat{Y}_t$	$e_t = Y_t - \hat{Y}_t$	$ e_t $	e_t^2	$ e_t/Y_t \times 100$
1	100	105	-5	5	25	5.00%
2	120	115	5	5	25	4.17%
3	110	112	-2	2	4	1.82%
4	130	125	5	5	25	3.85%
5	115	120	-5	5	25	4.35%
Sum/Mean	—	—	—	4.4	20.8	3.84%

MAE = 4.4, MSE = 20.8, RMSE = 4.56, MAPE = 3.84%

■ Interview Questions

- Q: What is RMSE and how do you interpret it? [RMSE = $\sqrt{\text{MSE}}$ = $\sqrt{\text{average squared error}}$. It measures average error in
- Q: When would you prefer MAE over RMSE? [When outliers are present and you don't want large errors to dominate. MAE
- Q: What is a disadvantage of MAPE? [It's undefined when $Y_t = 0$ (division by zero). Also, it's asymmetric — a 10% under-
- Q: If RMSE $\gg$ MAE for your model, what does that suggest? [There are some very large forecast errors (outliers) that are

CHAPTER 21

Time Series in the Wider Statistics Framework

Connecting time series to regression, probability, and econometrics

21.1 Time Series and Probability Theory

A time series $\{Y_t\}$ is a **stochastic process** — a collection of random variables indexed by time. Each Y_t is a random variable, and we observe ONE realisation of this process.

- Probability law: The full joint distribution $f(Y_1, Y_2, \dots, Y_T)$ characterises the process.
- Stationarity is a property of the stochastic PROCESS, not just one realisation.
- Ergodicity: the time average converges to the ensemble average (important for estimation).
- White noise $\varepsilon_t \sim \text{iid}(0, \sigma^2)$ is the building block — a pure random process.

21.2 Time Series vs Ordinary Regression

Feature	OLS Regression	Time Series Models
Independence assumption	Required: observations independent	Violated! Observations autocorrelated
Error term	$\varepsilon_t \sim \text{iid } N(0, \sigma^2)$	ε_t may be autocorrelated (need Durbin-Watson, etc.)
Regressor	Separate X variable	Lagged Y values (own past), or exogenous variables
OLS validity	Valid under independence	OLS is biased/inconsistent with lagged dependent variables
Time ordering	Not important	Critical — cannot shuffle observations
Spurious regression	Not a concern with $I(0)$ regressors	Serious concern with $I(1)$ non-stationary series

21.3 Spurious Regression — A Critical Warning

Spurious regression occurs when two **independently trending (non-stationary) series appear to be highly correlated**, even though they have NO real relationship. This is one of the biggest traps in time series analysis.

Classic example: UK shoe size and Indian GDP both trend upward. If you regress one on the other, you'll get $R^2=0.95$, $p<0.001$ — but it's completely meaningless!

Why it happens: Both series are $I(1)$ (non-stationary). Their trend causes the regression to look significant even with zero true relationship.

Solutions:

- Always test for stationarity before regression.
- If both series are $I(1)$: check for COINTEGRATION — if cointegrated, regression IS meaningful.
- If NOT cointegrated: difference both series first, then regress.

21.4 Time Series and Hypothesis Testing

Test	What it tests	H0	Decision rule
ADF Test	Unit root / stationarity	Series has unit root (non-stationary)	Reject H0 if $p < 0.05 \rightarrow$ STATIONARY
KPSS Test	Stationarity	Series IS stationary	Fail to reject H0 $\rightarrow$ STATIONARY
Ljung-Box Test	Residual autocorrelation	No autocorrelation in residuals (white noise)	Fail to reject H0 $\rightarrow$ GOOD model
Durbin-Watson	1st-order autocorrelation in residuals	No autocorrelation	$DW \approx 2 \rightarrow$ no autocorrelation
Breusch-Godfrey	Higher-order autocorrelation	No serial correlation up to order p	Reject H0 $\rightarrow$ autocorrelation present

21.5 Cointegration

Cointegration occurs when two or more $I(1)$ (non-stationary) series have a **long-run equilibrium relationship**. Although each series individually wanders (non-stationary), a linear combination of them is stationary.

Example: Stock prices of Reliance and HDFC both have unit roots individually. But the SPREAD (difference) between them might be stationary — they move together in the long run. This is cointegration, and it's economically meaningful.

If Y_t and X_t are both $I(1)$, but $(Y_t - \beta \cdot X_t)$ is $I(0) \rightarrow$ they are **COINTEGRATED**

Testing for Cointegration: Engle-Granger test or Johansen test.

■ Interview Questions

- Q: What is spurious regression? [When two unrelated non-stationary ($I(1)$) series appear to be correlated just because both are non-stationary]
- Q: What is cointegration? [Two $I(1)$ series are cointegrated if a linear combination of them is stationary. They have a long-run equilibrium relationship]
- Q: Why can't we use ordinary OLS for time series with lagged dependent variables? [Because the lagged dependent variable is correlated with the error term, violating the OLS assumptions]

CHAPTER 22

Advanced MSc Interview Topics

Unit roots, ADF, KPSS, cointegration, residual diagnostics

22.1 Unit Root — Deep Dive

A **unit root** exists when the characteristic equation of an AR process has a root equal to 1. For AR(1): $Y_t = \phi \cdot Y_{t-1} + \varepsilon_t$, the characteristic root is ϕ . If $\phi = 1 \rightarrow$ unit root $\rightarrow$ random walk $\rightarrow$ non-stationary.

AR(1): $\phi = 1 \rightarrow$ Unit Root $\rightarrow$ Non-Stationary (Random Walk)

AR(p): Characteristic equation $1 - \phi_1 B - \dots - \phi_p B^p = 0$ has root $B=1 \rightarrow$ Unit Root

Consequence: A series with a unit root is $I(1)$ — integrated of order 1. Differencing once gives a stationary series.

22.2 ADF Test — Augmented Dickey-Fuller

The **Augmented Dickey-Fuller (ADF) test** is the most widely used test for unit roots / stationarity.

The ADF regression:

$$\Delta Y_t = \alpha + \beta \cdot t + \gamma \cdot Y_{t-1} + \delta_1 \cdot \Delta Y_{t-1} + \dots + \delta_p \cdot \Delta Y_{t-p} + \varepsilon_t$$

Test the hypothesis: $H_0: \gamma = 0$ (unit root / non-stationary) vs $H_1: \gamma < 0$ (stationary)

Feature	Details
H_0 (null hypothesis)	$\gamma = 0 \rightarrow$ Series has a UNIT ROOT $\rightarrow$ NON-STATIONARY
H_1 (alternative)	$\gamma < 0 \rightarrow$ Series is STATIONARY (one-sided test)
Decision rule	If ADF test statistic < critical value (e.g. -2.86 at 5%): REJECT $H_0 \rightarrow$ Stationary
p-value decision	$p < 0.05$: Reject $H_0 \rightarrow$ STATIONARY. $p > 0.05$: Fail to reject $\rightarrow$ NON-STATIONARY
Critical values	Non-standard — must use Dickey-Fuller tables, NOT standard t-tables
'Augmented' meaning	Extra lagged differences $\Delta Y_{t-1}, \dots, \Delta Y_{t-p}$ added to remove serial correlation in residuals
Variants	Without constant, with constant, with constant+trend

■ Pro Tip

- CRITICAL MEMORY AID: ADF $p < 0.05 \rightarrow$ STATIONARY (reject unit root). This is the OPPOSITE of what many people expect.
- ADF rejects H_0 of NON-stationarity. So rejecting = stationary. This is a very common interview trap.

22.3 KPSS Test

The **KPSS (Kwiatkowski-Phillips-Schmidt-Shin) test** has the OPPOSITE null hypothesis to ADF.

Feature	ADF Test	KPSS Test
H_0 (null)	Series has unit root (NON-STATIONARY)	Series IS STATIONARY
H_1 (alternative)	Series is stationary	Series has unit root (non-stationary)
If $p < 0.05$	REJECT $H_0 \rightarrow$ Series IS stationary ■	REJECT $H_0 \rightarrow$ Series is NON-stationary ■
If $p > 0.05$	FAIL to reject $\rightarrow$ Non-stationary ■	FAIL to reject $\rightarrow$ Series IS stationary ■
Use together?	YES — use both for robust conclusion	YES — their opposite nulls complement each other

■ Pro Tip

- USE BOTH TESTS TOGETHER: If ADF rejects ($p < 0.05$) AND KPSS fails to reject ($p > 0.05$) $\rightarrow$ STRONG evidence of stationarity.
- If both tests agree on NON-stationarity $\rightarrow$ definitely non-stationary. If they conflict $\rightarrow$ inconclusive; use judgment.

22.4 Ljung-Box Test for Residual Autocorrelation

After fitting an ARIMA model, test if residuals are white noise:

$$Q_{LB} = T(T+2) \cdot \sum_{k=1}^m r_k^2 / (T-k)$$

where r_k is the sample autocorrelation of residuals at lag k , and m is the number of lags tested.

Under H_0 (residuals are white noise): $Q_{LB} \sim \chi^2(m-p-q)$

Result	Interpretation	Action
p-value > 0.05 (fail to reject H_0)	Cannot reject white noise $\rightarrow$ residuals look like white noise	Model is adequate — proceed to forecasting
p-value < 0.05 (reject H_0)	Residuals have significant autocorrelation ■	Model is inadequate $\rightarrow$ revise p or q

22.5 Seasonal Adjustment

Seasonal adjustment removes the seasonal component from a time series, leaving the trend and irregular components. Used by government agencies (e.g., seasonally-adjusted unemployment rate).

Methods:

- Census X-13 / X-12-ARIMA: Industry standard for economic data.
- STL decomposition (Seasonal and Trend decomposition using LOESS): More flexible.
- Classical decomposition: Simple but less accurate.

22.6 Integration Order — I(d) Notation

Notation	Meaning	Example
I(0)	Stationary — no differencing needed	White noise, AR(1) with $ \phi < 1$
I(1)	Needs 1 differencing to become stationary	Random walk, most economic series
I(2)	Needs 2 differencings to become stationary	Some price level series (rare)
I(d)	Needs d differencings to become stationary	General notation — 'd' in ARIMA(p,d,q)

■ Interview Questions

- Q: What is a unit root? [When the AR coefficient equals 1 ($\phi=1$), the series is a random walk — non-stationary with variable variance]
- Q: Explain the ADF test. What is H_0 ? [ADF tests $H_0: \gamma=0$ in $\Delta Y_t = \alpha + \beta t + \gamma Y_{t-1} + \dots$. A unit root exists under H_0 (non-stationary)]
- Q: How does KPSS differ from ADF? [KPSS has H_0 : series is STATIONARY (opposite of ADF). If KPSS fails to reject $\rightarrow$ series is stationary]
- Q: What is the Ljung-Box test? [Tests H_0 that first m autocorrelations of residuals are all zero (white noise). If $p > 0.05$, reject H_0]
- Q: What is I(1)? [A series is I(1) — 'integrated of order 1' — if it is non-stationary but becomes stationary after one differencing]

CHAPTER 23

150+ Interview Questions

Every question type an interviewer could ask — with model answers

PART A: Conceptual Questions (What is / Define)**Q1. What is a time series?**

📌 ■ A sequence of data points collected at successive points in time, usually at equally spaced intervals. Example: daily stock prices, monthly sales.

Q2. What are the four components of a time series?

📌 ■ Trend (T), Seasonality (S), Cyclical (C), Irregular (I). Remember: TSCI.

Q3. What is stationarity?

📌 ■ A series is weakly stationary if: (1) $E[Y_t] = \mu$ constant, (2) $\text{Var}(Y_t) = \sigma^2$ constant, (3) $\text{Cov}(Y_t, Y_{t+k}) = \gamma_k$ depends only on lag k , not on t .

Q4. What is the difference between weak and strict stationarity?

📌 ■ Weak: constant mean, variance, and autocovariance. Strict: entire joint distribution is time-invariant. Strict implies weak. For Gaussian processes they are equivalent.

Q5. What is white noise?

📌 ■ $E[\epsilon_t] = 0$, $\text{Var}(\epsilon_t) = \sigma^2$, $\text{Cov}(\epsilon_t, \epsilon_s) = 0$ for $t \neq s$. No autocorrelation, constant variance, zero mean. The building block of all time series models.

Q6. What is a random walk?

📌 ■ $Y_t = Y_{t-1} + \epsilon_t$. Non-stationary: variance grows with time ($t\sigma^2$). Best forecast = last value. It's AR(1) with $\phi=1$ (unit root).

Q7. What is the ACF?

📌 ■ Autocorrelation Function. $\rho_k = \text{Cov}(Y_t, Y_{t-k}) / \text{Var}(Y_t)$. Measures total correlation (direct + indirect) between Y_t and Y_{t-k} .

Q8. What is the PACF?

📌 ■ Partial Autocorrelation Function. ϕ_{kk} = direct correlation between Y_t and Y_{t-k} after removing the linear effects of intermediate lags.

Q9. What does ARIMA stand for?

📌 ■ AutoRegressive Integrated Moving Average. p =AR order, d =differencing, q =MA order. Handles non-stationary series.

Q10. What is differencing?

👉■ Computing $\Delta Y_t = Y_t - Y_{t-1}$. Removes trend and makes the series stationary. $d=1$ removes linear trend, $d=2$ removes quadratic.

Q11. What is an AR(p) model?

👉■ $Y_t = \phi_1 Y_{t-1} + \dots + \phi_p Y_{t-p} + \varepsilon_t$. Current value predicted from p past values. Stationarity requires $|\text{roots of AR polynomial}| > 1$.

Q12. What is an MA(q) model?

👉■ $Y_t = \varepsilon_t + \theta_1 \varepsilon_{t-1} + \dots + \theta_q \varepsilon_{t-q}$. Current value expressed through current and q past shocks. Always stationary.

Q13. What is exponential smoothing?

👉■ A forecasting method assigning exponentially decreasing weights to past observations. Most recent data gets highest weight. Parameter α controls the weight decay.

Q14. What is the Holt-Winters method?

👉■ Triple exponential smoothing for series with both trend and seasonality. Three parameters: α (level), β (trend), γ (seasonal).

Q15. What is SARIMA?

👉■ Seasonal ARIMA. $\text{SARIMA}(p,d,q)(P,D,Q)[s]$. Adds seasonal AR, differencing, and MA components at multiples of the seasonal period s .

PART B: Mathematical / Derivation Questions

Q16. Derive the mean and variance of AR(1): $Y_t = \phi Y_{t-1} + \varepsilon_t$.

👉■ By stationarity: $E[Y_t] = \mu$. Taking expectations: $\mu = \phi\mu \rightarrow \mu(1-\phi) = 0 \rightarrow \mu = 0$. For variance: $\text{Var}(Y_t) = \phi^2 \text{Var}(Y_{t-1}) + \sigma^2 = \phi^2 \sigma^2_Y + \sigma^2$. Since stationary: $\sigma^2_Y = \phi^2 \sigma^2_Y + \sigma^2 \rightarrow \sigma^2_Y = \sigma^2 / (1 - \phi^2)$.

Q17. What is the ACF of AR(1)?

👉■ $\rho_k = \phi^k$. Derived from: $\gamma_k = \text{Cov}(Y_t, Y_{t-k})$. Multiply both sides of AR(1) by Y_{t-k} and take expectations: $\gamma_k = \phi \gamma_{k-1}$. Dividing by γ_0 : $\rho_k = \phi \rho_{k-1} \rightarrow \rho_k = \phi^k$.

Q18. Derive the variance of MA(1): $Y_t = \varepsilon_t + \theta \varepsilon_{t-1}$.

👉■ $\text{Var}(Y_t) = \text{Var}(\varepsilon_t) + \theta^2 \text{Var}(\varepsilon_{t-1}) = \sigma^2 + \theta^2 \sigma^2 = \sigma^2(1 + \theta^2)$. (Uses independence of ε_t and ε_{t-1} .)

Q19. What is the autocorrelation of MA(1) at lag 1?

👉■ $\rho_1 = \text{Cov}(Y_t, Y_{t-1}) / \text{Var}(Y_t)$. $\text{Cov} = E[(\varepsilon_t + \theta \varepsilon_{t-1})(\varepsilon_{t-1} + \theta \varepsilon_{t-2})] = \theta \sigma^2$. So $\rho_1 = \theta \sigma^2 / \sigma^2(1 + \theta^2) = \theta / (1 + \theta^2)$.

Q20. What is the variance of a random walk after t steps?

👉■ $Y_t = \varepsilon_t + \varepsilon_{t-1} + \dots + \varepsilon_1$. $\text{Var}(Y_t) = t\sigma^2$ (since all ε_i are independent, variance adds up).

Q21. What is the Yule-Walker equation for AR(2)?

👉■ $\rho_1 = \phi_1 + \phi_2\rho_1 \rightarrow \phi_1 = \rho_1(1-\phi_2)/(1-\rho_1^2)$. And $\rho_2 = \phi_1\rho_1 + \phi_2$. These allow estimating ϕ_1, ϕ_2 from sample ACF.

Q22. What is the stationarity condition for AR(2)?

👉■ Three conditions must all hold: (1) $\phi_1 + \phi_2 < 1$, (2) $\phi_2 - \phi_1 < 1$, (3) $|\phi_2| < 1$. Or equivalently: roots of $1 - \phi_1 B - \phi_2 B^2 = 0$ lie OUTSIDE the unit circle.

Q23. What are the 95% confidence bands for an ACF plot?

👉■ $\pm 1.96/\sqrt{T}$, where T is the number of observations. For T=100: ± 0.196 . Bars exceeding this bound are statistically significant at the 5% level.

PART C: Application / Practical Questions

Q24. How would you approach a time series forecasting problem from scratch?

👉■ 1) Plot and explore data. 2) Check stationarity (ADF test). 3) Transform/difference if needed. 4) Plot ACF/PACF. 5) Identify p,d,q. 6) Fit ARIMA, estimate parameters. 7) Check residuals (Ljung-Box). 8) Forecast with prediction intervals. 9) Evaluate on test set (MAE, RMSE, MAPE).

Q25. You see the ACF of a series decays very slowly. What do you do?

👉■ Slow ACF decay signals NON-STATIONARITY. Apply first differencing, then re-plot the ACF of the differenced series. Do NOT attempt to identify p or q before achieving stationarity.

Q26. ACF cuts off after lag 2. PACF tails off. What model do you fit?

👉■ MA(2). The ACF cutoff after q=2 with PACF tailing off is the diagnostic signature of an MA(2) process.

Q27. PACF cuts off after lag 1. ACF tails off. Model?

👉■ AR(1). The PACF cutoff at p=1 with ACF tailing off identifies AR(1).

Q28. Both ACF and PACF tail off. What do you do?

👉■ This suggests an ARMA(p,q) model. Try multiple combinations: ARMA(1,1), ARMA(2,1), ARMA(1,2), etc. Select using AIC or BIC (lower is better).

Q29. Your ARIMA residuals show significant ACF at lag 5. What does this mean?

👉■ The model has not fully captured the autocorrelation structure. The residuals are NOT white noise. Consider increasing p or q, or checking if d was chosen correctly.

Q30. How would you forecast monthly airline passengers? What model?

👉■ Airline passengers show trend AND multiplicative seasonal pattern. Steps: 1) Take log to stabilise variance. 2) Apply seasonal differencing (D=1, s=12). 3) Apply regular differencing (d=1). 4) Identify P,Q from ACF/PACF at seasonal lags. 5) Fit SARIMA(0,1,1)(0,1,1)[12] — the classic airline model.

PART D: Tricky / Twisted Interview Questions

Q31. Is white noise stationary? Justify formally.

👉■ YES. Check all three conditions: (1) $E[\varepsilon_t]=0$ (constant). (2) $\text{Var}(\varepsilon_t)=\sigma^2$ (constant). (3) $\text{Cov}(\varepsilon_t, \varepsilon_s)=0$ for $t \neq s$, so autocovariance depends only on lag (it's zero for all non-zero lags). All three conditions satisfied $\rightarrow$ weakly stationary.

Q32. Is a random walk stationary? Can its mean be constant?

👉■ TRICKY! The MEAN of a random walk $Y_0=0$ is $E[Y_t]=0$ (constant!). But it is still NOT stationary because $\text{Var}(Y_t)=t\sigma^2$ grows with t . Stationarity requires BOTH constant mean AND constant variance.

Q33. Can you have ARMA(0,0)? What is it?

👉■ Yes! ARMA(0,0) means no AR and no MA terms $\rightarrow Y_t = \varepsilon_t \rightarrow$ White Noise. It's a valid special case.

Q34. Why is ARIMA(0,1,0) a random walk?

👉■ $d=1$ means $\Delta Y_t = Y_t - Y_{t-1}$ is modelled, with $p=0, q=0$, so $\Delta Y_t = \varepsilon_t$ (white noise). Therefore $Y_t = Y_{t-1} + \varepsilon_t \rightarrow$ random walk.

Q35. If you difference a stationary series, what happens?

👉■ You OVER-DIFFERENCE. The ACF will show a significant negative spike at lag 1. You've introduced artificial negative autocorrelation that wasn't originally there. Always test stationarity BEFORE differencing.

Q36. Why can't we use a regular t-table for ADF test critical values?

👉■ Because under H_0 (unit root), the test statistic does NOT follow a standard t-distribution. The distribution of the ADF statistic was derived by Dickey and Fuller using Monte Carlo simulation. Non-standard distribution, so use Dickey-Fuller tables.

Q37. ARIMA(0,1,1) is equivalent to what forecasting method?

👉■ Simple Exponential Smoothing (SES). This is a famous result by Box and Jenkins. The α parameter in SES corresponds to $1+\theta$ in the ARIMA(0,1,1) model.

Q38. Can MA models have infinite memory?

👉■ No! MA(q) has FINITE memory of exactly q periods. Past shocks ε_{t-k} for $k > q$ have NO effect on Y_t . In contrast, AR models have INFINITE but decaying memory.

Q39. What if the AR coefficient is exactly -1?

👉■ AR(1) with $\phi=-1$: $Y_t = -Y_{t-1} + \varepsilon_t$. The series alternates in sign endlessly and has growing variance — non-stationary. We need $|\phi| < 1$ strictly.

Q40. What is the minimum number of parameters to model $Y_t = 0.7Y_{t-1} + \varepsilon_t + 0.4\varepsilon_{t-1}$?

👉■ ARMA(1,1): just 2 parameters ($\phi_1=0.7$, $\theta_1=0.4$). If you used a pure AR to represent this, you'd need infinite AR lags. ARMA is more parsimonious.

PART E: Follow-up / Cross-Questions

Q41-50: Follow-up chains

- Q41: You said AR(1) is stationary when $|\phi| < 1$. What happens at the boundary $\phi=1$? [Unit root $\rightarrow$ random walk $\rightarrow$ non-stationary. This is the critical boundary case.]

- Q42: You fitted an AR(2). The Ljung-Box test fails. What next? [The residuals aren't white noise. Re-examine ACF/PACF of residuals, try increasing p or adding MA terms. Possibly try ARMA(2,1) or ARMA(2,2).]
- Q43: You said MAPE can't be used when $Y_t=0$. What do you use instead? [Use MAE or RMSE (no division by actual values). Or use MASE (Mean Absolute Scaled Error) which also avoids division-by-zero issues.]
- Q44: You used first differencing ($d=1$). How do you convert the forecasts back to the original scale? [Inverse differencing: $\hat{Y}_{T+1} = Y_T + \Delta \hat{Y}_{T+1}$. For multiple differences, apply the inverse operation iteratively.]
- Q45: Your ADF test says non-stationary, but KPSS says stationary. What do you conclude? [The tests are conflicting — inconclusive result. Look at the series visually, check the test statistics more carefully. Often indicates near-unit-root behaviour or structural breaks.]
- Q46: You have monthly data. How many seasonal differences might you need? [Usually $D=1$ (one seasonal difference at $s=12$) is sufficient. Check ACF at lags 12, 24 after seasonal differencing.]
- Q47: What is parsimony and why does it matter in model selection? [Parsimony means using the simplest model (fewest parameters) that adequately fits the data. Complex models overfit training data and perform poorly on new data. AIC and BIC penalise complexity.]
- Q48: What is the difference between in-sample and out-of-sample forecast accuracy? [In-sample: accuracy on the data used to fit the model (always looks better, can be misleading). Out-of-sample: accuracy on held-out test data (true measure of predictive performance).]
- Q49: Two models have AIC of 150 and 145 respectively. Which do you prefer? [The model with AIC=145 is preferred — lower AIC means better balance between fit and parsimony. A difference of 5 is substantial (>2 is considered meaningful).]
- Q50: Can you forecast further than $h=1$ without increasing uncertainty? [No. Forecast uncertainty ALWAYS increases with horizon h . This is a fundamental property — the further ahead, the wider the prediction intervals.]

PART F: Professor-Level Questions

- Q51: What is the Wold decomposition theorem? [Any covariance-stationary process can be decomposed into a deterministic part and an infinite MA(∞) part: $Y_t = \sum \psi_j \cdot \varepsilon_{t-j}$. Justifies why MA representation is fundamental to time series.]
- Q52: Why do we use MLE for ARIMA estimation rather than OLS? [ARIMA models have MA components and lagged dependent variables. OLS is inconsistent in these cases. MLE properly accounts for the joint distribution of the data and gives consistent, efficient estimates.]
- Q53: What is a cointegrated system? What is an Error Correction Model (ECM)? [If two I(1) series are cointegrated, the Granger Representation Theorem says they can be modelled with an ECM: $\Delta Y_t = \alpha(Y_{t-1} - \beta X_{t-1}) + \text{lagged terms} + \varepsilon_t$. The ' α ' term is the error correction mechanism — it corrects deviations from long-run equilibrium.]

- Q54: What is GARCH and when would you need it? [ARCH/GARCH models volatility (conditional variance) of financial series. $GARCH(1,1)$: $\sigma^2_t = \omega + \alpha \cdot \varepsilon^2_{t-1} + \beta \cdot \sigma^2_{t-1}$. Used when residuals show ARCH effects — 'volatility clustering' (large shocks followed by large shocks).]
- Q55: What is the difference between a deterministic and stochastic trend? [Deterministic trend: $Y_t = \beta_0 + \beta_1 \cdot t + \varepsilon_t$ — trend is a fixed function of t . Remove by detrending (OLS). Stochastic trend: $Y_t = Y_{t-1} + \varepsilon_t$ — random walk type. Remove by differencing. CRITICAL distinction!]

PART G: Numerical MCQ-Style Questions

- Q56: For $AR(1)$ with $\phi=0.5$, what is $Var(Y_t)$? [$\sigma^2/(1-0.25) = \sigma^2/0.75 = 1.333\sigma^2$]
- Q57: MAPE for actuals {100,200} and forecasts {110,180}? [$|10|/100 \times 100 = 10\%$, $|20|/200 \times 100 = 10\%$. MAPE=10%]
- Q58: ACF at lag 0 is always what? [$\rho_0 = 1.0$ (always)]
- Q59: For $T=196$ data points, what are the 95% ACF confidence bounds? [$\pm 1.96/\sqrt{196} = \pm 1.96/14 = \pm 0.14$]
- Q60: For $MA(1)$ with $\theta=0.8$, what is ρ_1 ? [$\theta/(1+\theta^2) = 0.8/1.64 = 0.488$]
- Q61: Random walk: $Y_0=5$, $\varepsilon_1=2$, $\varepsilon_2=-1$, $\varepsilon_3=3$. What is Y_3 ? [$Y_3=Y_0+\varepsilon_1+\varepsilon_2+\varepsilon_3=5+2-1+3=9$]
- Q62: SES with $\alpha=0.4$. $S_1=100$, $Y_2=110$. What is S_2 ? [$S_2=0.4 \times 110 + 0.6 \times 100 = 44 + 60 = 104$]
- Q63: What is the minimum number of data points before the 5-period SMA can be computed? [5 (need at least 5 observations for the first SMA value at $t=5$)]
- Q64: $ARIMA(2,1,1)$ — how many non-seasonal parameters total? [$p+q = 2+1 = 3$ parameters (ϕ_1 , ϕ_2 , θ_1), plus possibly a constant]
- Q65: For a 3-point MA, how many observations are 'lost' at each end? [1 at each end = 2 total. $k=3$, so $(k-1)/2 = 1$.]

PART H: Quick-Fire 30-Second Answer Questions

- Q66: Define lag. [A lag k means Y_{t-k} : the value of the series k time periods before t .]
- Q67: What does 'integrated of order 1' ($I(1)$) mean? [The series is non-stationary but becomes stationary after one differencing.]
- Q68: What is forecast bias? [Systematic error in one direction — consistently over- or under-forecasting. Measured by mean error: $(1/n)\sum(Y_t - \hat{Y}_t)$.]
- Q69: What is a correlogram? [Another name for the ACF plot. It plots autocorrelations ρ_k against lag k .]
- Q70: What is invertibility? [An $MA(q)$ model is invertible if it can be rewritten as an infinite AR. Ensures unique parameter estimates. Condition for $MA(1)$: $|\theta| < 1$.]
- Q71: What is the seasonal period for monthly data with annual seasonality? [$s = 12$]

- Q72: What does $d=2$ in ARIMA mean? [The series was differenced TWICE to achieve stationarity — it had a quadratic or accelerating trend.]
- Q73: What is the backshift operator B ? [$B^k Y_t = Y_{t-k}$. It 'shifts' the series back by k periods.]
- Q74: Is SARIMA(0,0,0)(0,0,0)[12] meaningful? [It's equivalent to white noise — no structure at all. Rarely used but valid.]
- Q75: What is an in-sample forecast? [A forecast made for observations WITHIN the training data period — used to assess model fit, not true forecast accuracy.]

CHAPTER 24

Complete Interview Cheat Sheet

Everything you need — one page revision for the night before the interview

ALL DEFINITIONS — Quick Reference

Term	Definition (10 words or less)
Time Series	Data collected sequentially over time at equal intervals
Trend (T)	Long-term upward or downward direction in data
Seasonality (S)	Regular, fixed-period repeating pattern (within a year)
Cyclical (C)	Wave-like fluctuation over 2–10 years (variable period)
Irregular (I)	Unpredictable random residual — cannot be modelled
Stationarity	Constant mean, variance, and autocovariance over time
White Noise	Zero mean, constant variance, zero autocorrelation
Random Walk	$Y_t = Y_{t-1} + \epsilon_t$. Non-stationary. Unit root.
ACF at lag k	Total correlation between Y_t and Y_{t-k}
PACF at lag k	Direct correlation between Y_t and Y_{t-k} (controlling lags 1...k-1)
Differencing	$\Delta Y_t = Y_t - Y_{t-1}$. Removes trend. Makes series stationary.
AR(p)	$Y_t = \sum \phi_i Y_{t-i} + \epsilon_t$. Past values predict current.
MA(q)	$Y_t = \epsilon_t + \sum \theta_i \epsilon_{t-i}$. Past shocks drive current value.
ARMA(p,q)	Combination of AR(p) and MA(q). For stationary series.
ARIMA(p,d,q)	ARMA after d-fold differencing. For non-stationary series.
SARIMA	(p,d,q)(P,D,Q)[s]. ARIMA extended for seasonal data.
Invertibility	MA model can be rewritten as infinite AR. Requires $ \text{roots} > 1$.
Unit Root	AR coefficient = 1. Series is non-stationary (random walk).
ADF Test	Tests H_0 : unit root. Reject ($p < 0.05$) → STATIONARY.
KPSS Test	Tests H_0 : stationary. Reject ($p < 0.05$) → NON-STATIONARY.
Ljung-Box	Tests H_0 : residuals are white noise. Want $p > 0.05$ (good model).
Cointegration	Two I(1) series have stable long-run linear relationship.
Spurious Regression	Two unrelated trending series appear correlated. Dangerous!
MAE	Mean Absolute Error = $(1/n) \sum e_t $. Same units as Y.
MSE	Mean Squared Error = $(1/n) \sum e_t^2$. Penalises large errors heavily.

RMSE	Root MSE = $\sqrt{\text{MSE}}$. Same units as Y. Most common metric.
MAPE	Mean Abs Percentage Error = $(100/n)\sum e_t/Y_t $. Scale-free.

ALL KEY FORMULAS

AR(1): $Y_t = \phi Y_{t-1} + \varepsilon_t$. Mean=0. $\text{Var}=\sigma^2/(1-\phi^2)$. ACF: $\rho_k=\phi^k$

AR(p): $Y_t = \phi_1 Y_{t-1} + \dots + \phi_p Y_{t-p} + \varepsilon_t$. Stationary if $|\text{roots}|>1$.

MA(1): $Y_t = \varepsilon_t + \theta \varepsilon_{t-1}$. $\text{Var}=\sigma^2(1+\theta^2)$. $\rho_1=\theta/(1+\theta^2)$. $\rho_k=0$ ($k\geq 2$).

MA(q): $Y_t = \varepsilon_t + \theta_1 \varepsilon_{t-1} + \dots + \theta_q \varepsilon_{t-q}$. ACF cuts off after lag q. Always stationary.

ARMA(1,1): $Y_t = \phi Y_{t-1} + \varepsilon_t + \theta \varepsilon_{t-1}$. Need $|\phi|<1$ (stationary) AND $|\theta|<1$ (invertible).

ARIMA(p,d,q): $\Phi(B)(1-B)^d Y_t = \Theta(B)\varepsilon_t$. First difference d times, then ARMA.

SARIMA: $(p,d,q)(P,D,Q)[s]$: $\Phi(B)\Phi_s(B^s)(1-B)^d(1-B^s)^D Y_t = \Theta(B)\Theta_s(B^s)\varepsilon_t$

SES: $S_t = \alpha Y_t + (1-\alpha)S_{t-1}$. Forecast = S_t .

Holt's Method: $L_t = \alpha Y_t + (1-\alpha)(L_{t-1} + T_{t-1})$. $T_t = \beta(L_t - L_{t-1}) + (1-\beta)T_{t-1}$.

ACF: $\rho_k = \text{Cov}(Y_t, Y_{t-k})/\text{Var}(Y_t) = \gamma_k/\gamma_0$

PACF lag 2: $\phi_{22} = (\rho_{22} - \rho_{12}^2)/(1 - \rho_{12}^2)$

95% ACF Bands: $\pm 1.96/\sqrt{T}$

ADF Test (H0): $\gamma=0$ in $\Delta Y_t = \alpha + \beta t + \gamma Y_{t-1} + \sum \delta_i \Delta Y_{t-i} + \varepsilon$. Reject if $p < 0.05 \rightarrow$ stationary.

Ljung-Box: $Q = T(T+2)\sum r_k^2/(T-k)$. Under H_0 : $\sim \chi^2(m-p-q)$. Want $p > 0.05$.

MAE: $(1/n)\sum |Y_t - \hat{Y}_t|$

RMSE: $\sqrt{[(1/n)\sum (Y_t - \hat{Y}_t)^2]}$

MAPE: $(100/n)\sum |Y_t - \hat{Y}_t|/|Y_t|$

AIC: $-2\ln(L) + 2k$. Lower is better.

BIC: $-2\ln(L) + k \cdot \ln(T)$. Penalises parameters more than AIC.

1st Difference: $\Delta Y_t = Y_t - Y_{t-1}$

2nd Difference: $\Delta^2 Y_t = Y_t - 2Y_{t-1} + Y_{t-2}$

Seasonal Diff: $\Delta_s Y_t = Y_t - Y_{t-s}$

Var(RW at t): $\text{Var}(Y_t) = t\sigma^2$ (grows without bound)

AR(1) Forecast: $\hat{Y}_{T+h} = \phi^h \cdot Y_T$ (reverts to mean as $h \rightarrow \infty$)

THE GOLDEN RULES — Must Memorise

■ GOLDEN RULE 1: ACF CUTS OFF after lag q $\rightarrow$ MA(q). ACF TAILS OFF $\rightarrow$ AR or ARMA.

- GOLDEN RULE 2: PACF CUTS OFF after lag $p \rightarrow \text{AR}(p)$. PACF TAILS OFF $\rightarrow$ MA or ARMA.
- GOLDEN RULE 3: Both ACF and PACF TAIL OFF $\rightarrow \text{ARMA}(p,q)$. Use AIC/BIC.
- GOLDEN RULE 4: SLOW ACF DECAY $\rightarrow$ NON-STATIONARY. Difference first. NEVER identify p,q on non-stationary series.
- GOLDEN RULE 5: ADF $p < 0.05 \rightarrow$ STATIONARY (reject unit root). KPSS $p < 0.05 \rightarrow$ NON-STATIONARY (reject stationarity).
- GOLDEN RULE 6: After model fitting, check RESIDUALS = white noise (Ljung-Box $p > 0.05$).
- GOLDEN RULE 7: $\text{ARIMA}(0,1,1) \equiv \text{SES}$. $\text{ARIMA}(0,2,2) \equiv$ Holt's Linear Trend. Classic equivalences!
- GOLDEN RULE 8: MA models are ALWAYS stationary. AR models need $|\phi| < 1$ for stationarity.
- GOLDEN RULE 9: Seasonal data $\rightarrow \text{SARIMA}(p,d,q)(P,D,Q)[s]$. $s=12$ monthly, $s=4$ quarterly.
- GOLDEN RULE 10: $d=1$ removes linear trend. $d=2$ removes quadratic trend. Rarely need $d \geq 3$.

COMMON TRAPS — Never Fall For These

- TRAP 1: 'ACF cuts off at lag 2 $\rightarrow \text{AR}(2)$.' WRONG! ACF cutoff $\rightarrow \text{MA}(2)$. PACF cutoff $\rightarrow \text{AR}$.
- TRAP 2: 'Random walk is stationary because its mean is zero.' WRONG! Variance grows $\rightarrow$ not stationary.
- TRAP 3: 'We should always difference to make data stationary.' WRONG! If series is already stationary ($I(0)$), differencing makes it WORSE (over-differencing).
- TRAP 4: 'ADF $p > 0.05$ means stationary.' WRONG! $p > 0.05$ means FAIL to reject H_0 of NON-STATIONARITY.
- TRAP 5: 'Seasonal and cyclical mean the same thing.' WRONG! Seasonal = FIXED period (≤ 1 year). Cyclical = VARIABLE period (years).
- TRAP 6: 'MA(q) is like a q -period moving average of Y .' WRONG! MA(q) is a moving average of ERRORS, not of Y values.
- TRAP 7: 'High R^2 in regression means two time series are related.' Not if both are non-stationary! This could be spurious regression.
- TRAP 8: 'Prediction intervals and confidence intervals are the same.' WRONG! PI is for future OBSERVATION (wider). CI is for PARAMETER (narrower).
- TRAP 9: 'ARIMA works for seasonal data.' Standard ARIMA does NOT handle seasonality. Need SARIMA.
- TRAP 10: 'Strict stationarity is what we use.' Almost always mean WEAK (covariance) stationarity in practice.

30-SECOND CHEAT SHEET — ACF/PACF to Model

Situation	Model	Action
-----------	-------	--------

ACF slow decay	Non-stationary	DIFFERENCE first!
ACF cuts off at q, PACF tails off	MA(q)	Fit MA(q)
PACF cuts off at p, ACF tails off	AR(p)	Fit AR(p)
Both tail off	ARMA(p,q)	Try (1,1), (2,1), (1,2) — use AIC
All near zero	White noise	No model needed
Spikes at lag s, 2s in ACF	Seasonality	Add D=1, seasonal MA (Q)
After fitting: residual ACF has spikes	Bad model	Increase p or q, re-check d
After fitting: residual ACF all clear	Good model!	Forecast!

BOX-JENKINS 4-STEP SUMMARY

Step	Action	Key tool
1. Identify	Plot data, test stationarity, difference if needed, read ACF/PACF	ADF test, ACF, PACF plots
2. Estimate	Fit ARIMA with chosen p,d,q. Estimate ϕ 's and θ 's	Maximum Likelihood (MLE)
3. Diagnose	Check if residuals are white noise	Ljung-Box test, residual ACF
4. Forecast	Generate predictions with prediction intervals, evaluate	MAE, RMSE, MAPE on test set

■ You Are Now Interview-Ready!

You have covered all 24 chapters: from zero to MSc Statistics interview level.

Time Series Analysis — Complete Study Guide | All Rights Reserved